

TIME-VARYING TRANSITION PROBABILITY MATRIX

APPLICATION TO BRAND SHARE ANALYSIS

Noboru Murata

June 8, 2024

<https://noboru-murata.github.io/>

Introduction

- motivated examples

- stationary distribution of Google matrix

Problem Formulation

- time-varying graph and transition matrix

- graph estimation problem

Numerical Examples

- real-world data analysis

Conclusion

INTRODUCTION

two problems in graph inference:

- **direct problem (deductive):**

induce properties from known graph structure

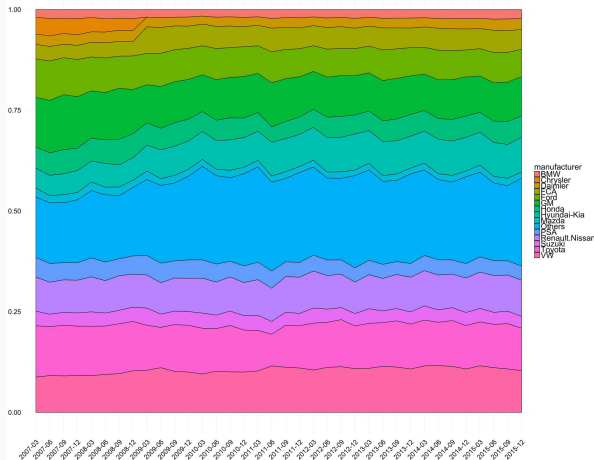
- shortest path, traveling salesman, graph coloring problem
- maximum flow, minimum cut problem
- Google search (stationary distribution of Google matrix)

- **inverse problem (inductive):**

estimate graph structure from partially observed properties

- path analysis, graphical model estimation
- sparse estimation of accuracy matrix
- structure estimation via graph Laplacian

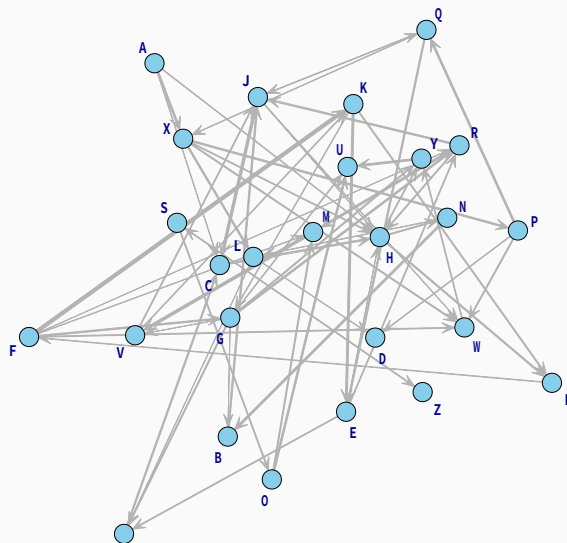
automobile sales data of manufacturers



Questions

- why sales shares vary?
- what happens in customer preferences?

weighted graph



directed graph

- number of node: 26
- edge exist.: 0.1
- weight:
uniform on $[0, 1]$

- adjacency matrix W

$(W)_{ij}$ = strength of connection from i to j

- indicate vector of sink node $\mathbf{a}$

$$(\mathbf{a})_i = \begin{cases} 1, & \text{if } (W\mathbf{e})_i = 0 \\ 0, & \text{otherwise} \end{cases}$$

where $\mathbf{e}$ is a vector of all 1

- normalized adjacency matrix (transition matrix) H

$$H = \text{diag}(W\mathbf{e} + \mathbf{a})^{-1}W$$

$(H\mathbf{e})_i = 1$ holds except for sink nodes

- transition matrix with sink node adjustment S

$$S = H + \mathbf{a}\mathbf{e}^T/n$$

= (probabilistic transition)
+ (escape from sink nodes)

probability matrix: $S\mathbf{e} = \mathbf{e}$

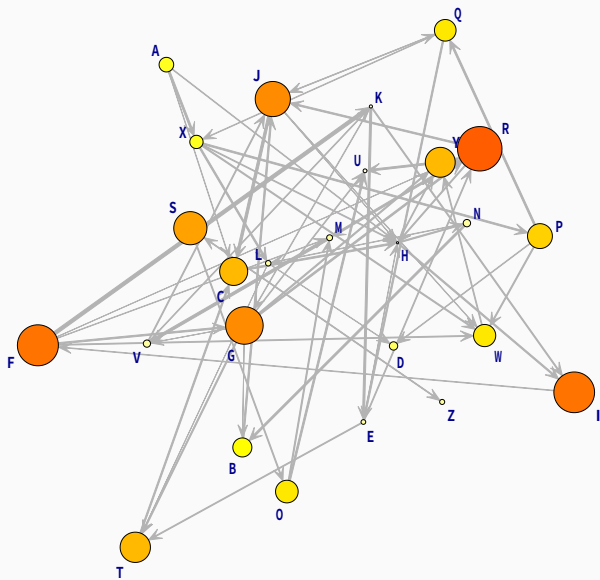
- Google matrix G

$$G = \alpha S + (1 - \alpha)\mathbf{e}\mathbf{e}^T/n$$

= (transition along edges) + (random transition)

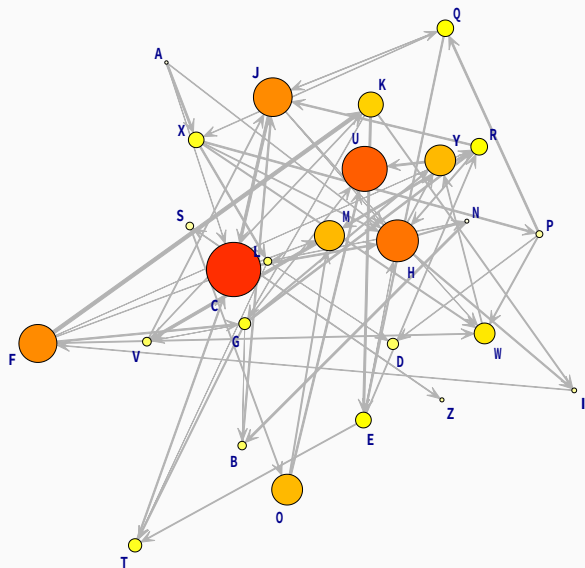
probability matrix: $G\mathbf{e} = \mathbf{e}$

t = 1



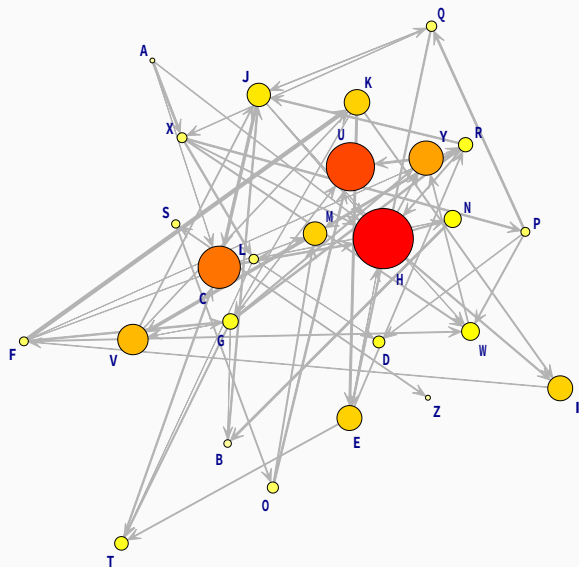
- $\alpha = 0.85$
- initial: uniform dist.

$t = 2$



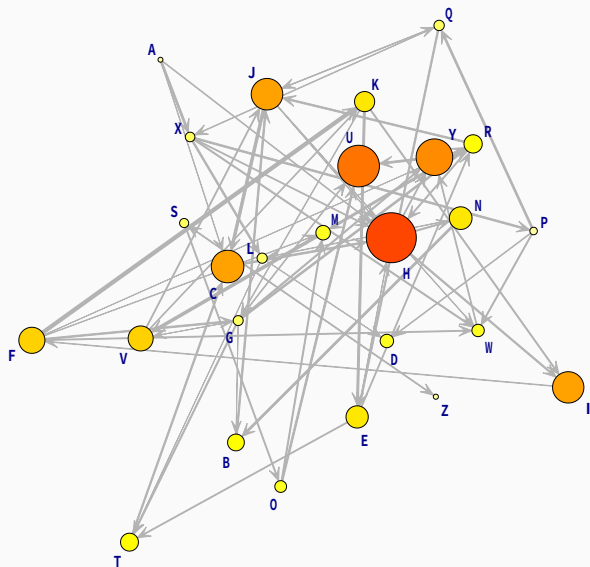
- $\alpha = 0.85$
- initial: uniform dist.

t = 3



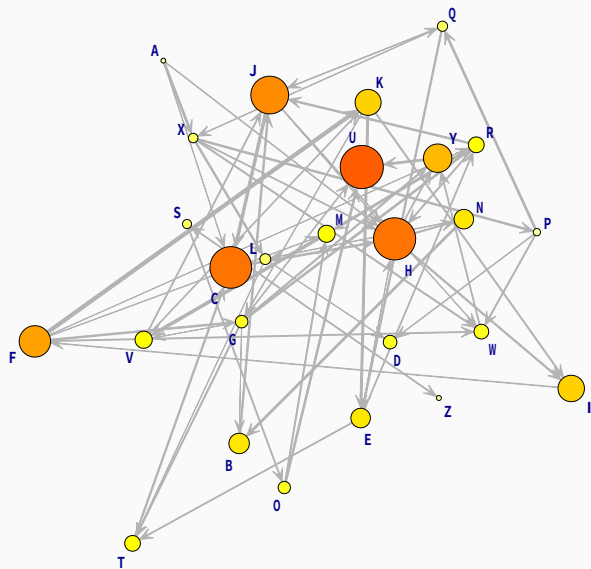
- $\alpha = 0.85$
- initial: uniform dist.

t = 4



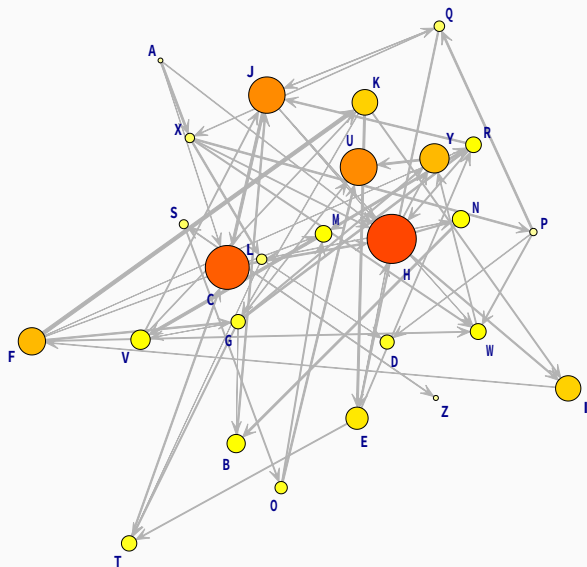
- $\alpha = 0.85$
- initial: uniform dist.

t = 5



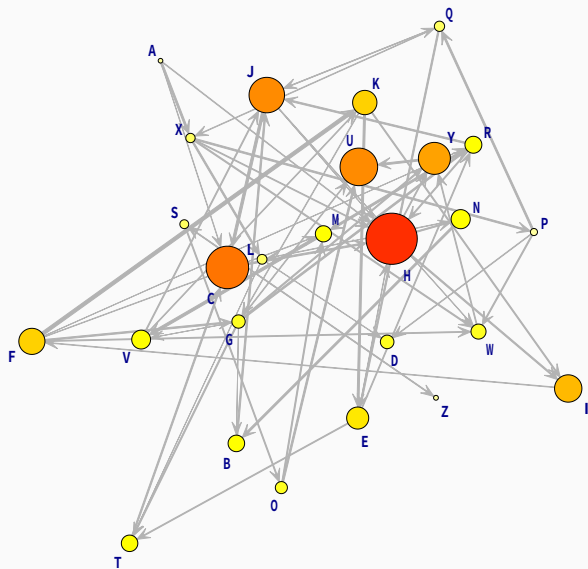
- $\alpha = 0.85$
- initial: uniform dist.

t = 6



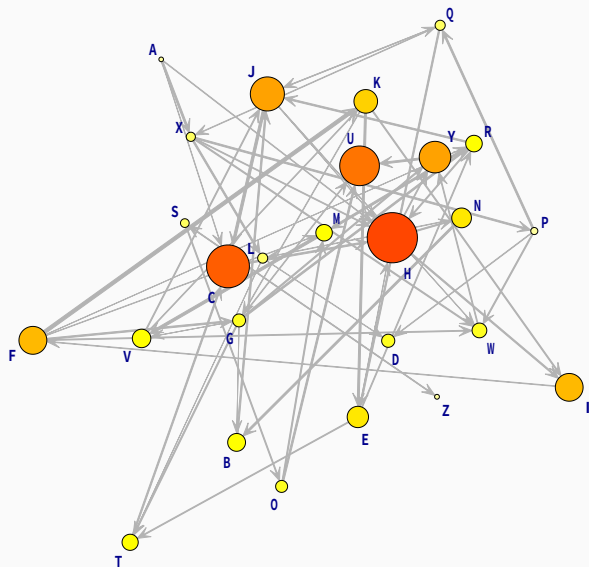
- $\alpha = 0.85$
- initial: uniform dist.

t = 7



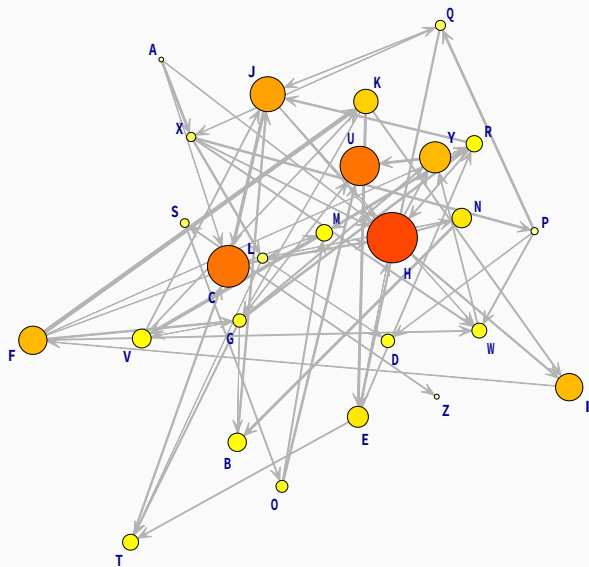
- $\alpha = 0.85$
- initial: uniform dist.

t = 8



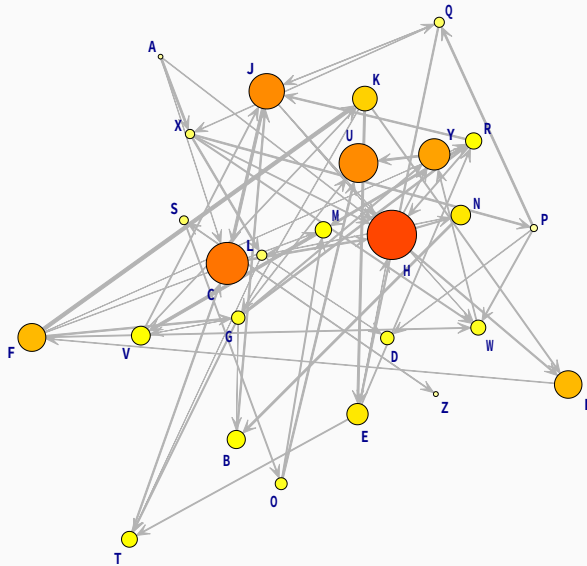
- $\alpha = 0.85$
- initial: uniform dist.

t = 9



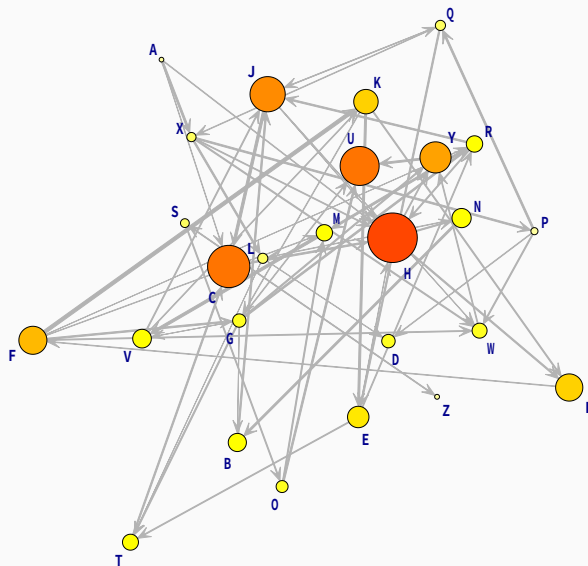
- $\alpha = 0.85$
- initial: uniform dist.

t = 10



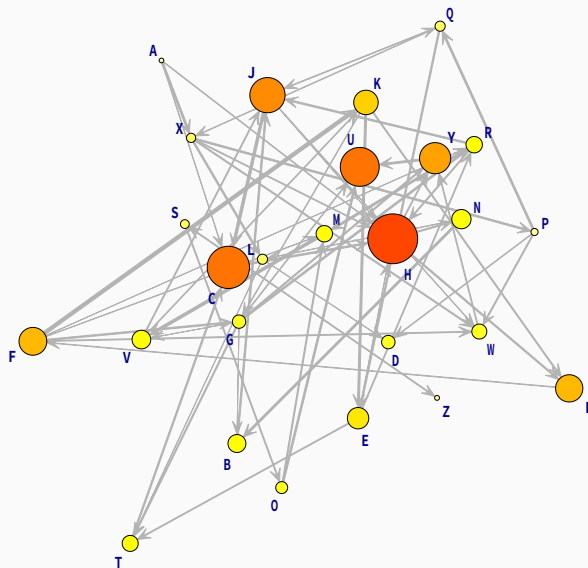
- $\alpha = 0.85$
- initial: uniform dist.

t = 11



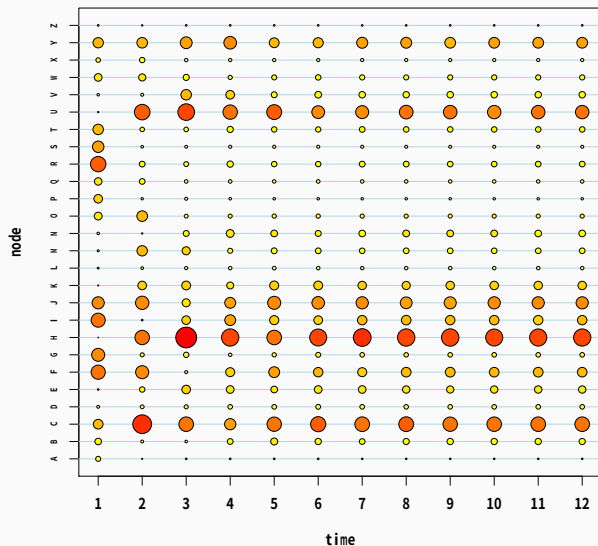
- $\alpha = 0.85$
- initial: uniform dist.

t = 12



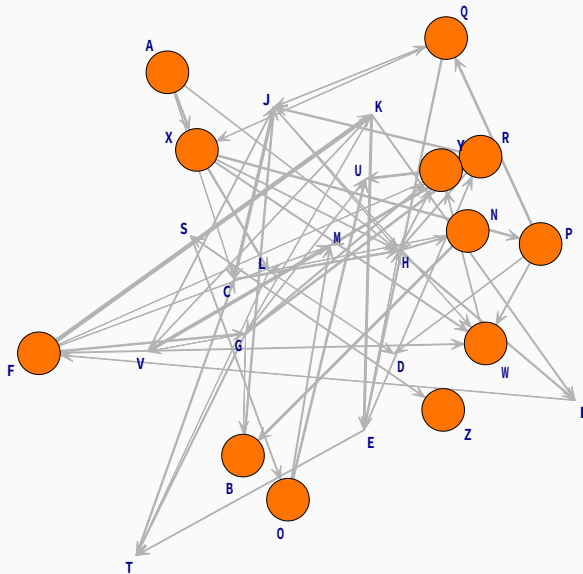
- $\alpha = 0.85$
- initial: uniform dist.

transient process



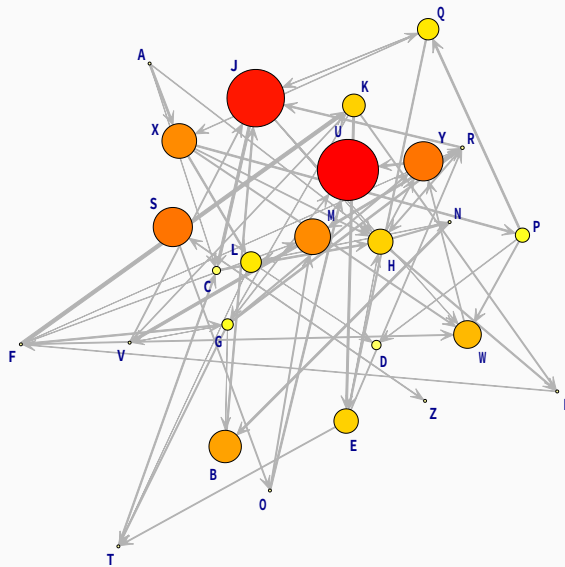
- $\alpha = 0.85$
- initial: uniform dist.

t = 1



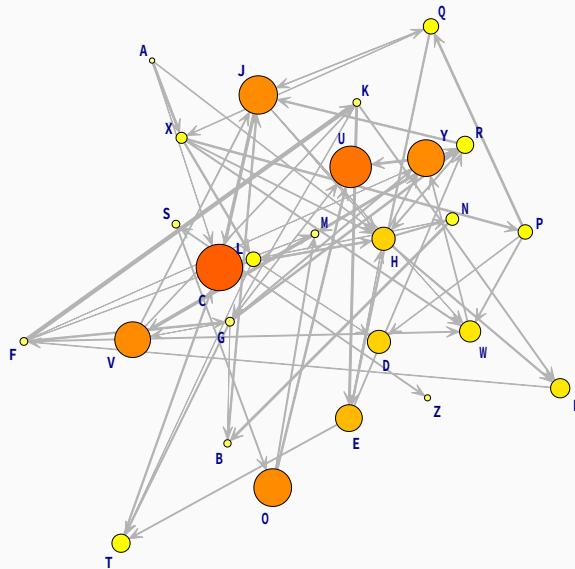
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 2



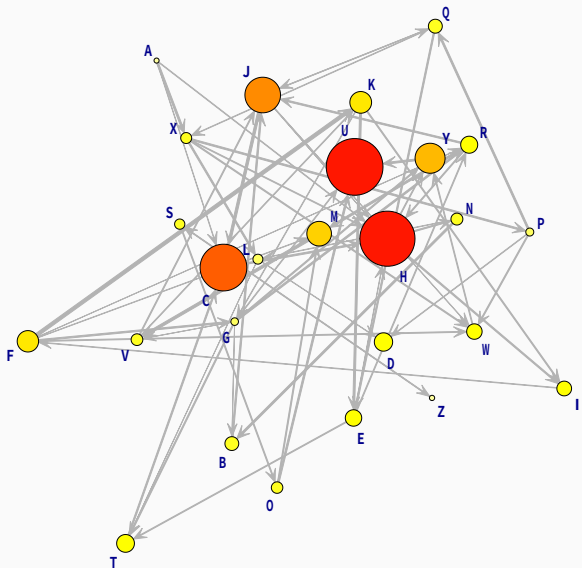
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 3



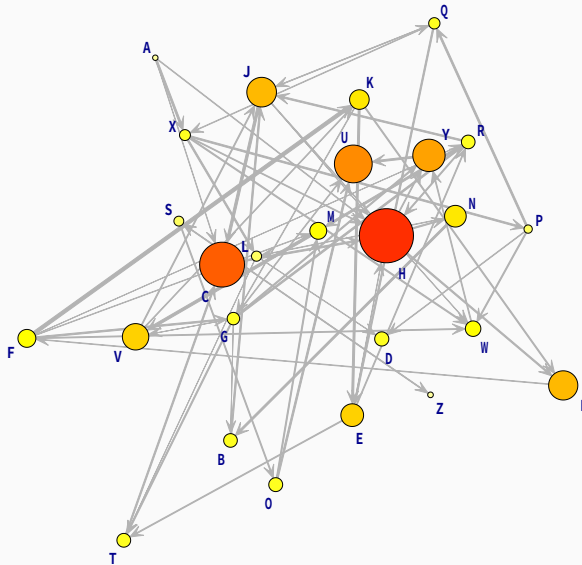
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 4



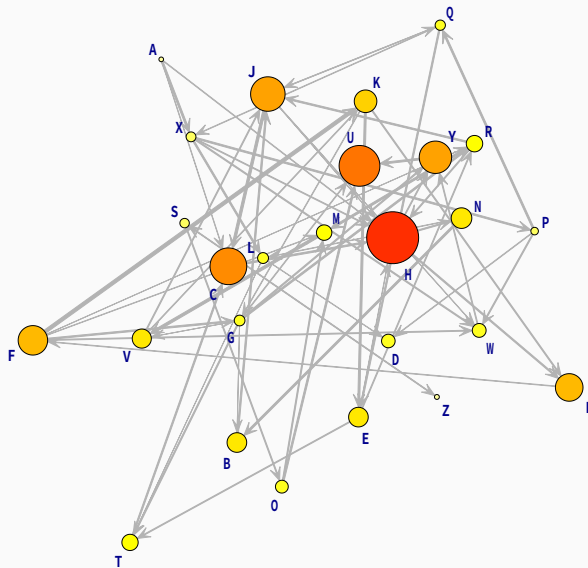
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 5



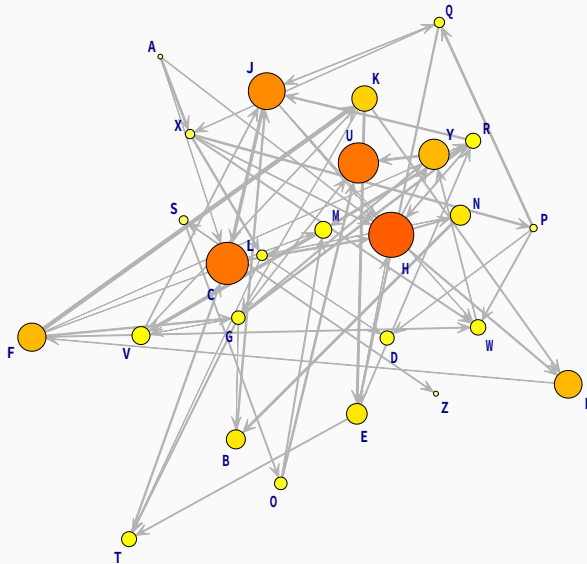
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

$t = 6$



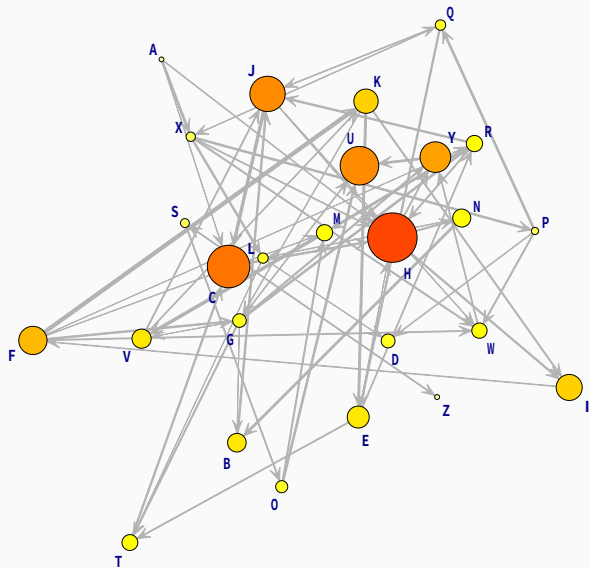
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 7



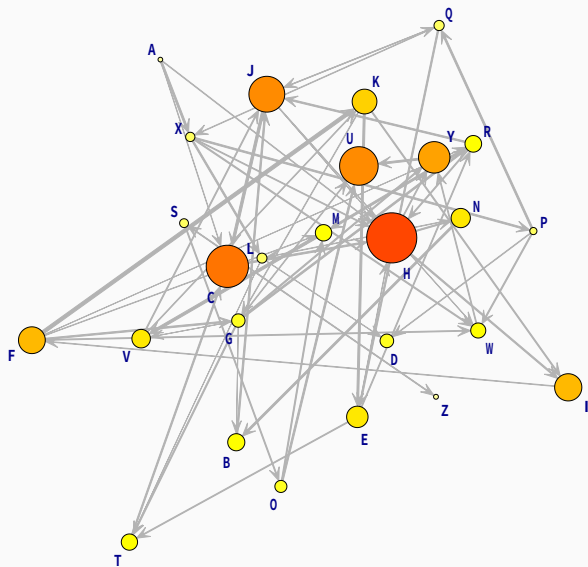
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 8



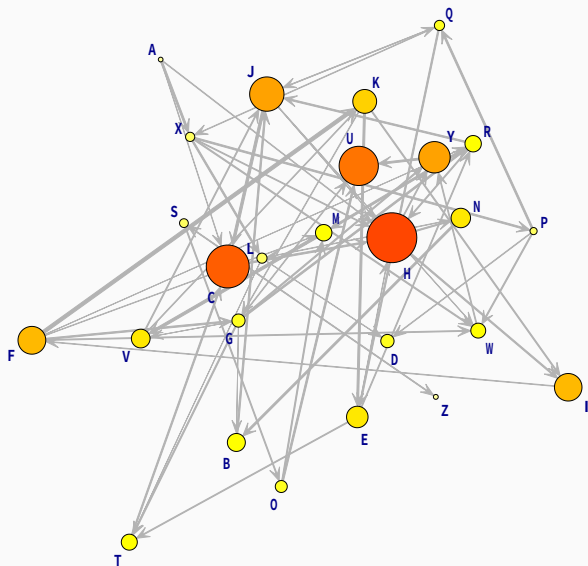
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 9



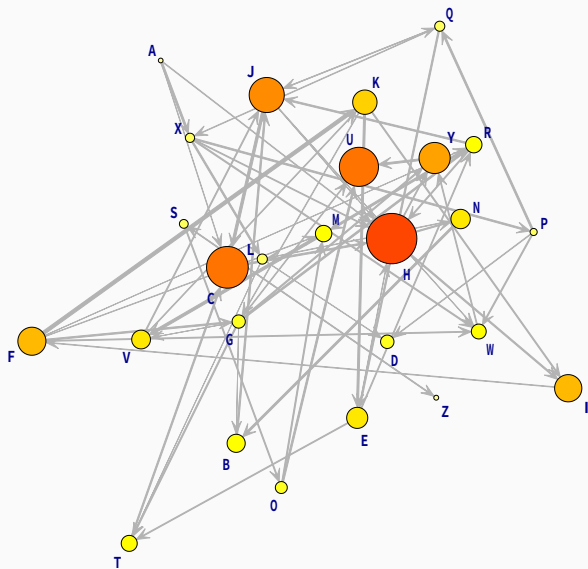
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 10



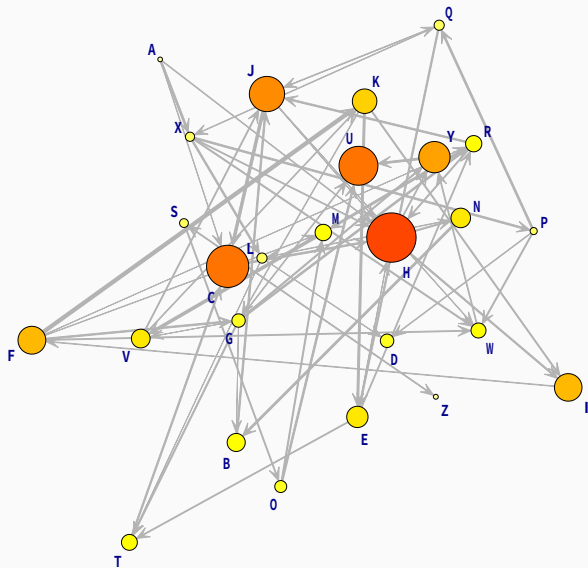
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 11



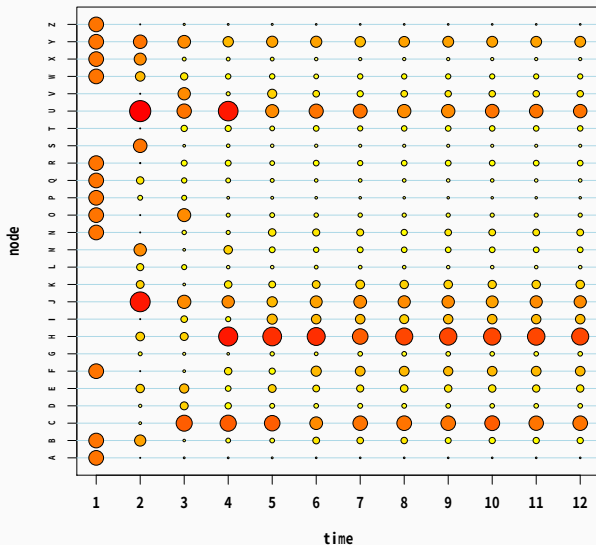
- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

t = 12



- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

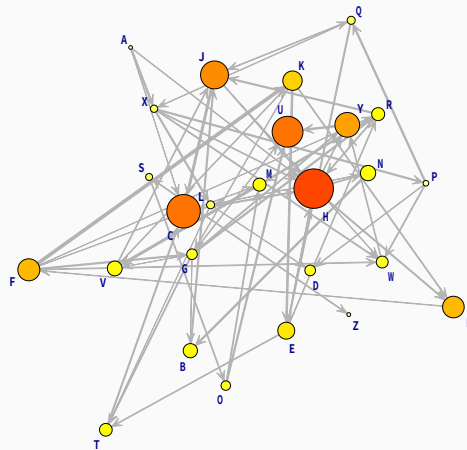
transient process



- $\alpha = 0.85$
- initial: 12 nodes (randomly selected)

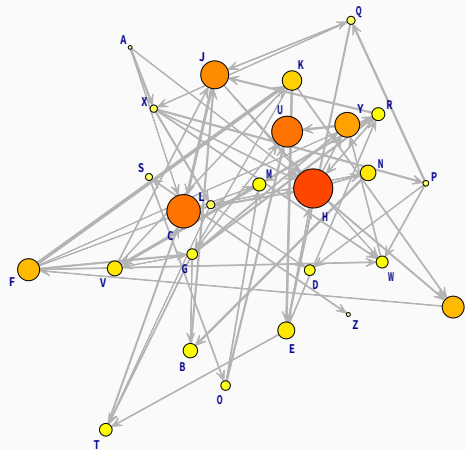
initial condition 1

$t = 12$



initial condition 2

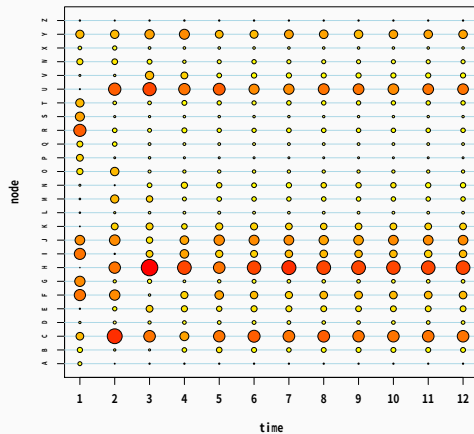
$t = 12$



fast convergence to the same stationary distribution

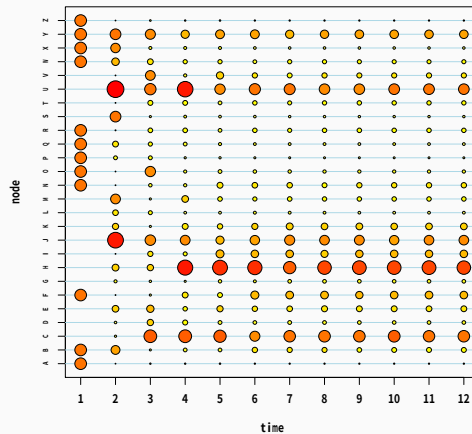
initial condition 1

transient process



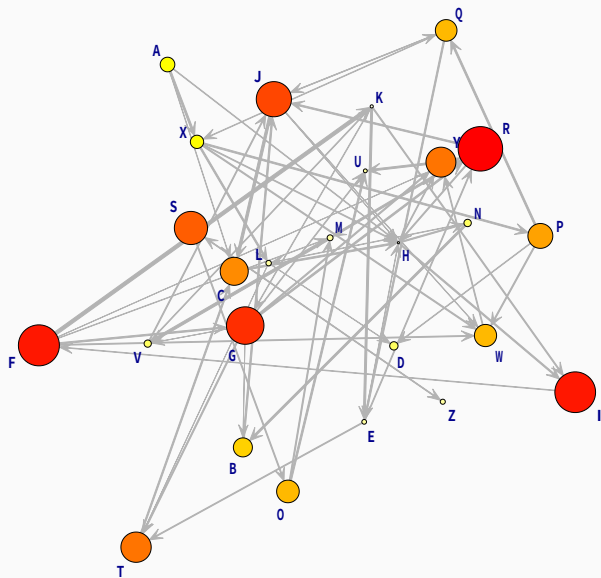
initial condition 2

transient process



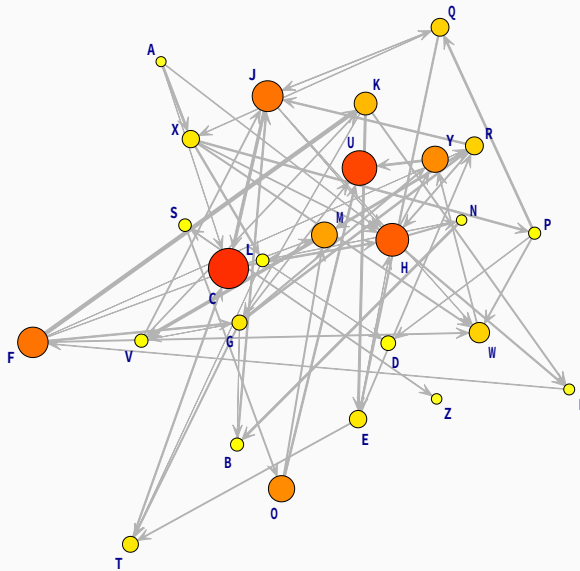
fast convergence to the same stationary distribution

t = 1



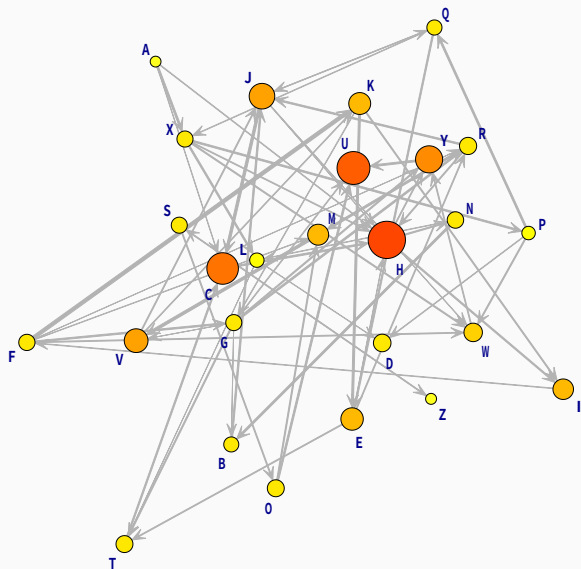
- $\alpha = 0.5$
- initial: uniform dist.

t = 2



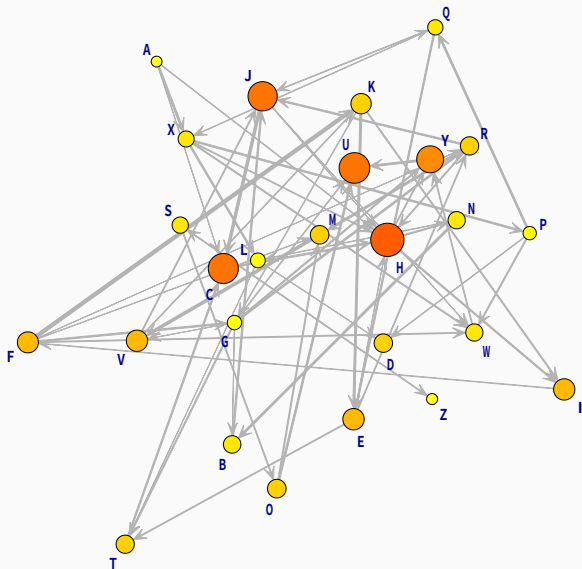
- $\alpha = 0.5$
- initial: uniform dist.

t = 3



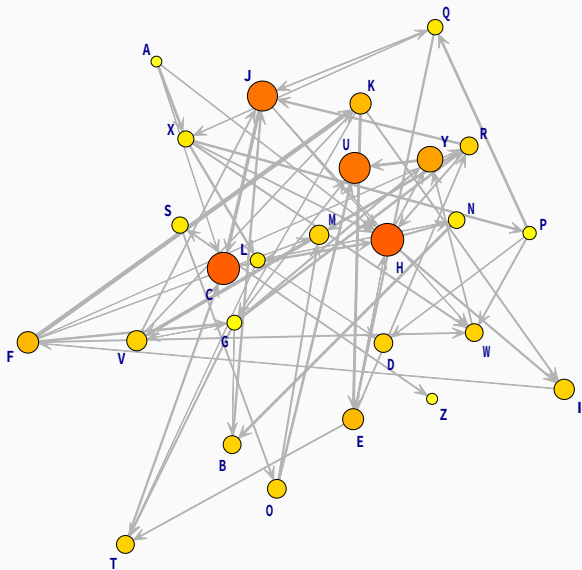
- $\alpha = 0.5$
- initial: uniform dist.

t = 4



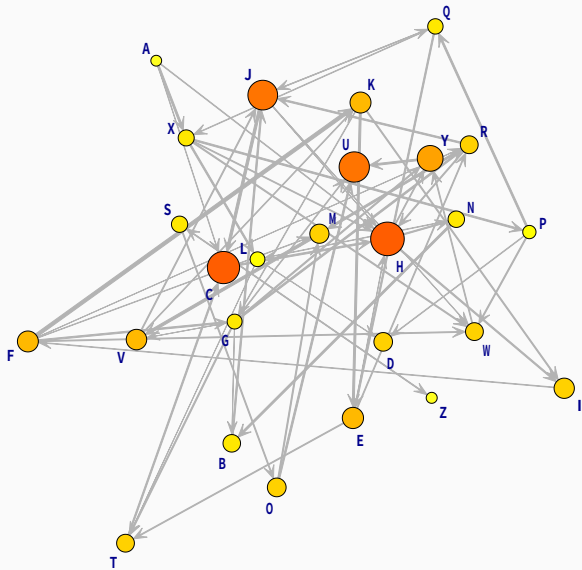
- $\alpha = 0.5$
- initial: uniform dist.

t = 5



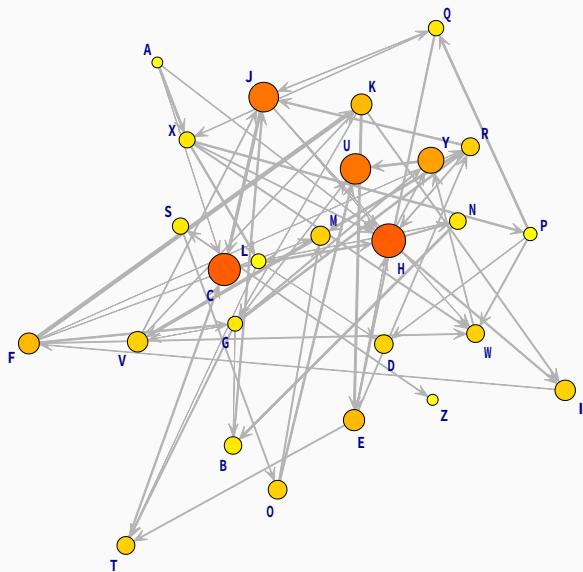
- $\alpha = 0.5$
- initial: uniform dist.

t = 6



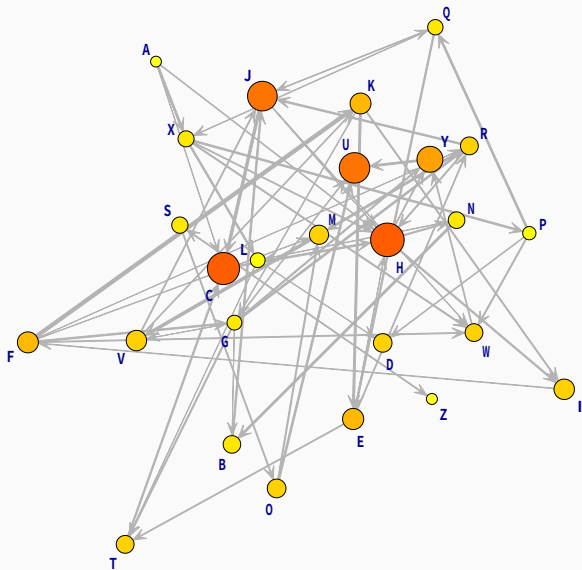
- $\alpha = 0.5$
- initial: uniform dist.

t = 7



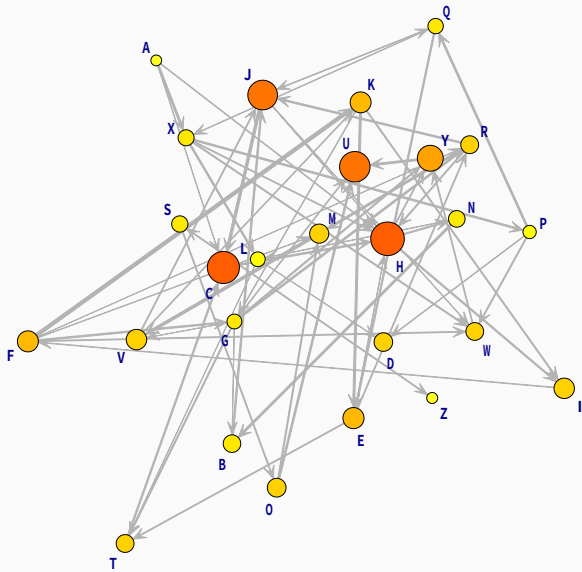
- $\alpha = 0.5$
- initial: uniform dist.

t = 8



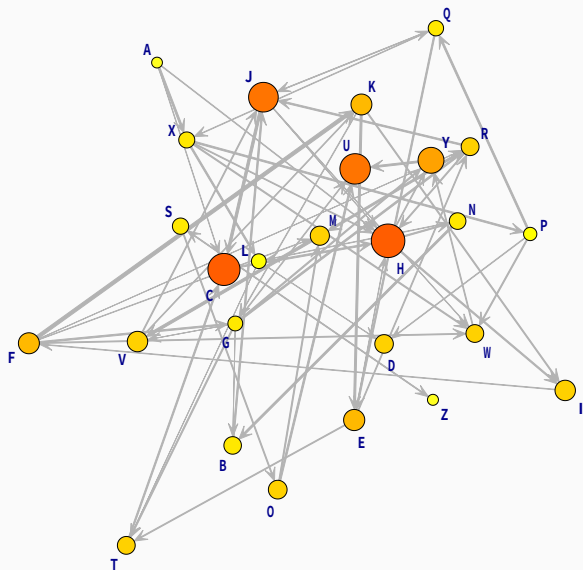
- $\alpha = 0.5$
- initial: uniform dist.

t = 9



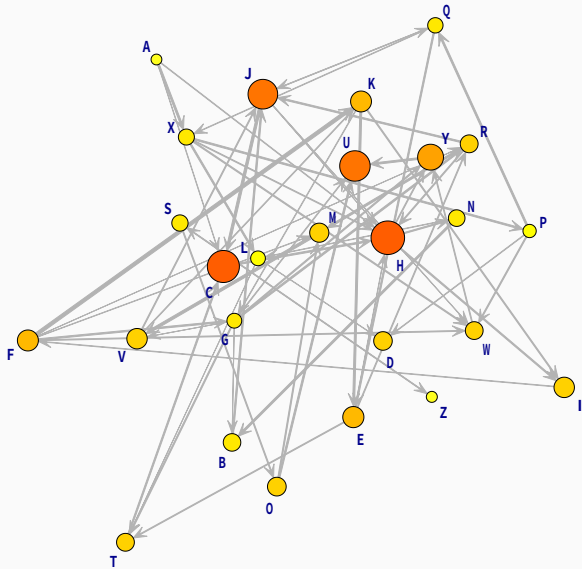
- $\alpha = 0.5$
- initial: uniform dist.

t = 10



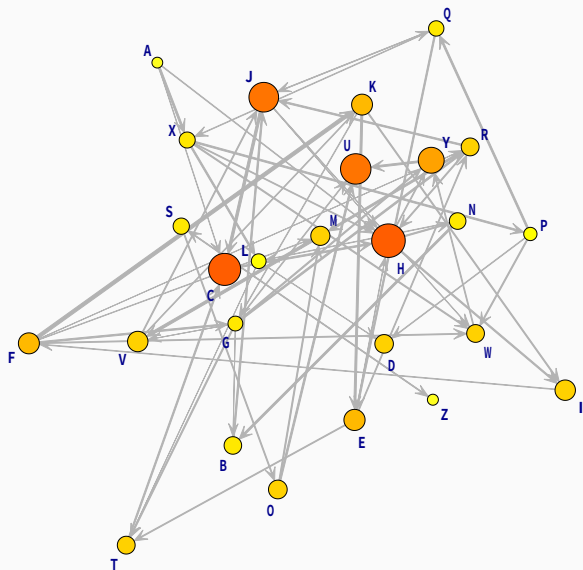
- $\alpha = 0.5$
- initial: uniform dist.

t = 11



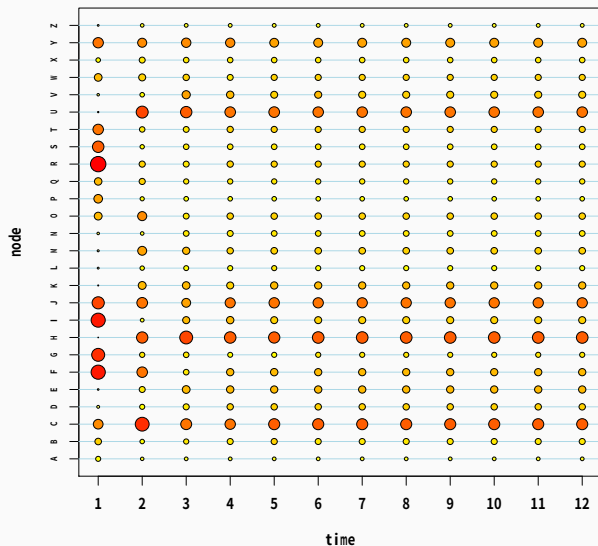
- $\alpha = 0.5$
- initial: uniform dist.

t = 12



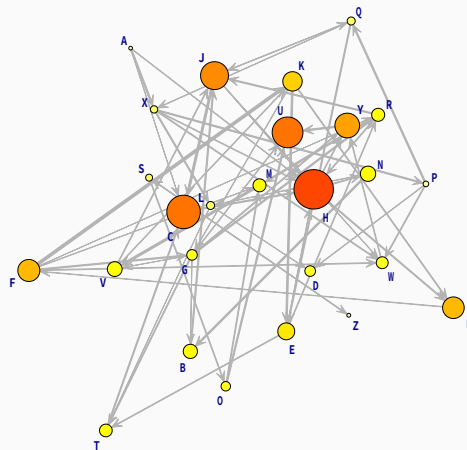
- $\alpha = 0.5$
- initial: uniform dist.

transient process

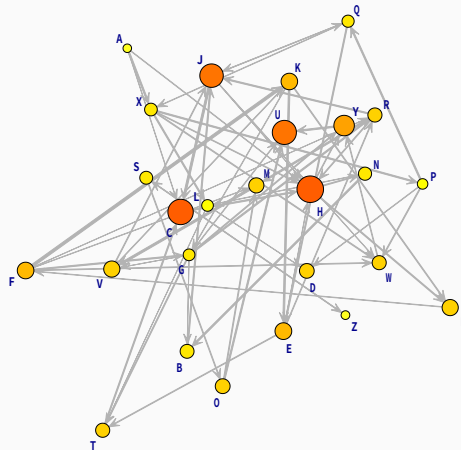


- $\alpha = 0.5$
- initial: uniform dist.

$\alpha = 0.85$
 $t = 12$



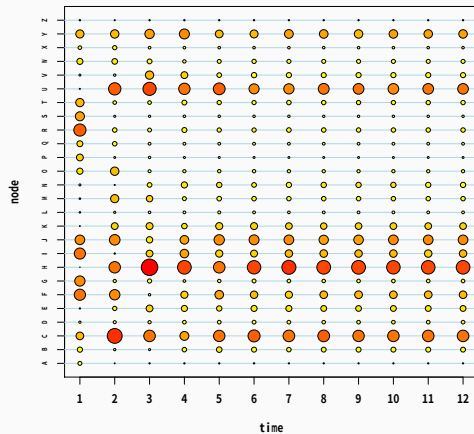
$\alpha = 0.5$
 $t = 12$



random diffusion with smaller scaling

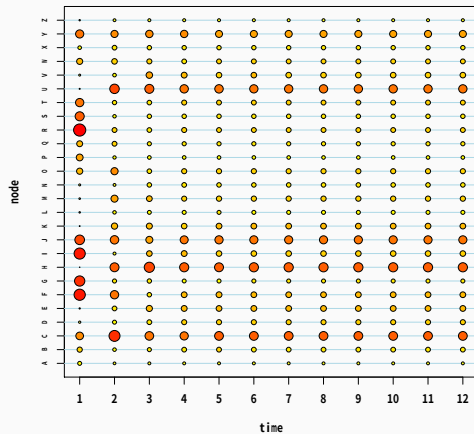
$$\alpha = 0.85$$

transient process



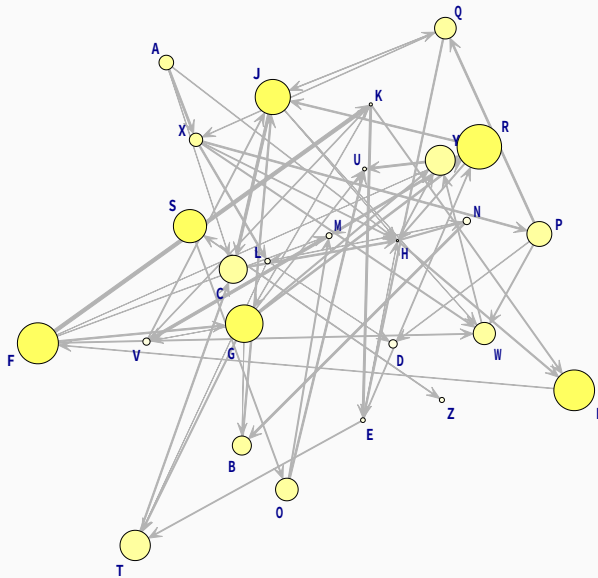
$$\alpha = 0.5$$

transient process



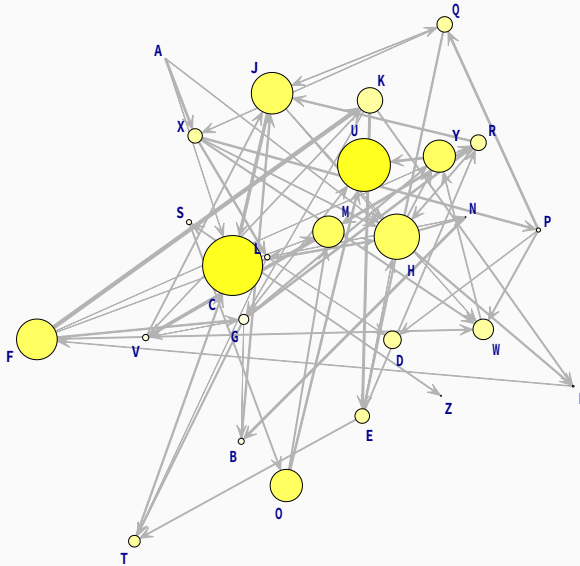
random diffusion with smaller scaling

t = 1



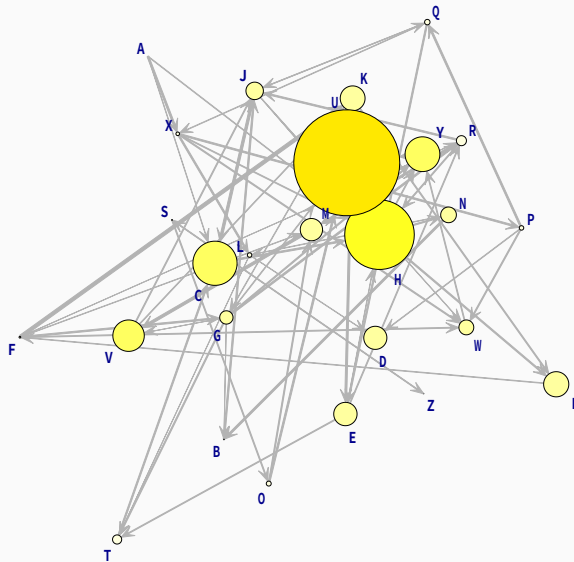
- without sink node escape
- initial: uniform dist.

$t = 2$



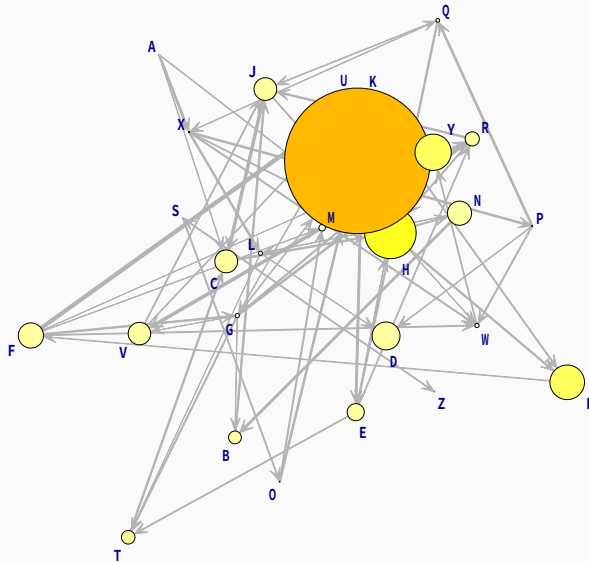
- without sink node escape
- initial: uniform dist.

t = 3



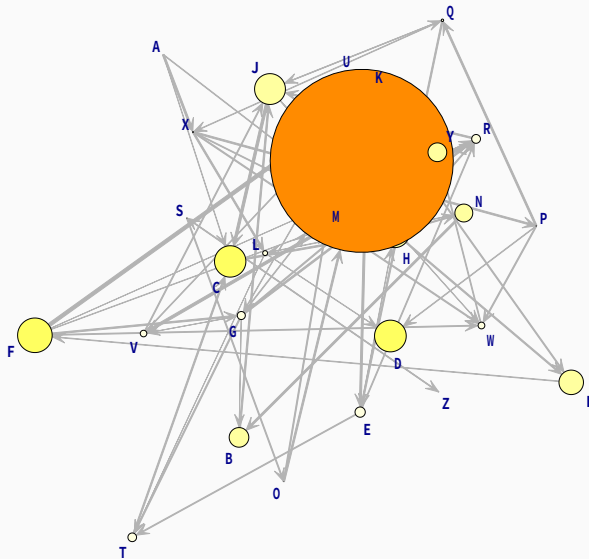
- without sink node escape
- initial: uniform dist.

$t = 4$



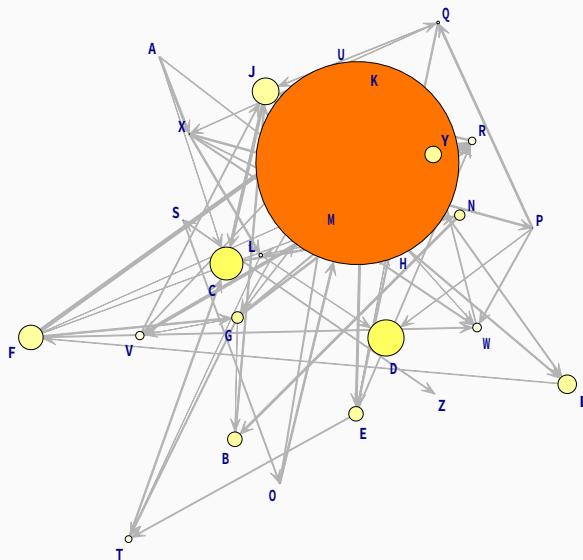
- without sink node escape
- initial: uniform dist.

$t = 5$



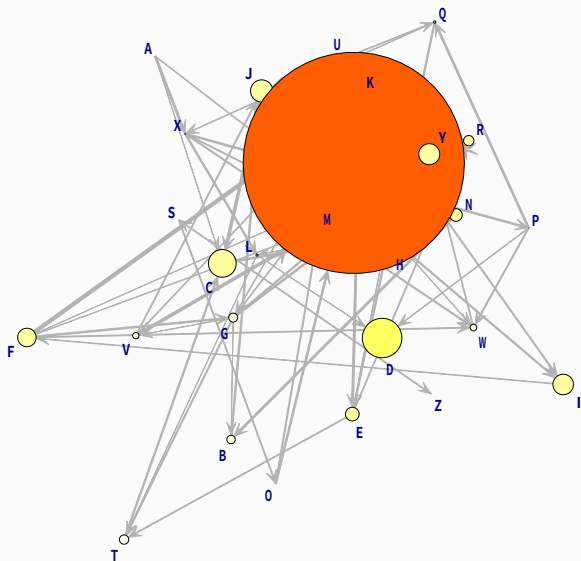
- without sink node escape
- initial: uniform dist.

$t = 6$



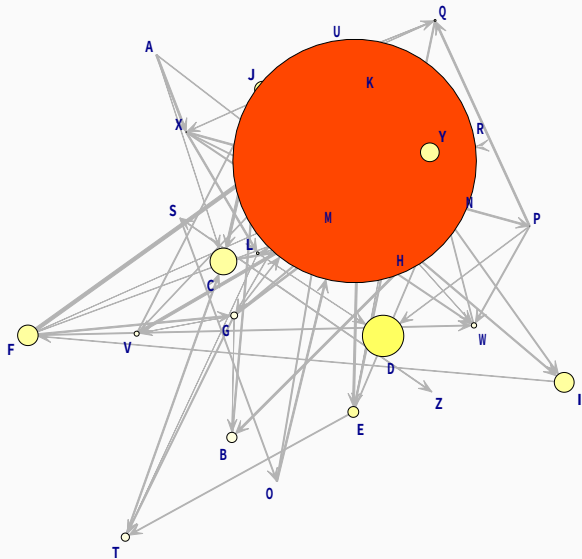
- without sink node escape
- initial: uniform dist.

t = 7



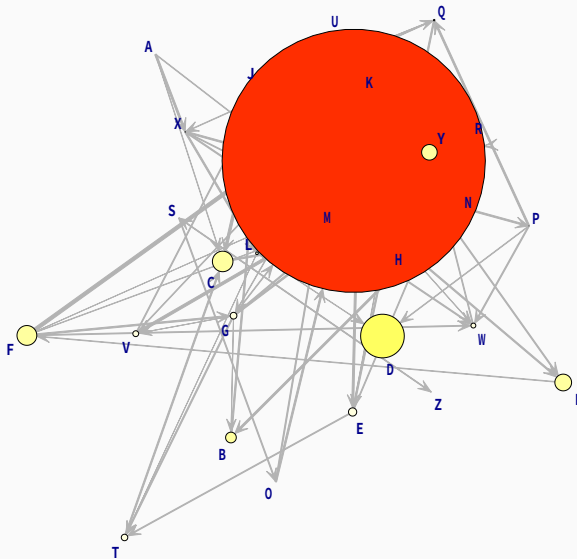
- without sink node escape
- initial: uniform dist.

$t = 8$



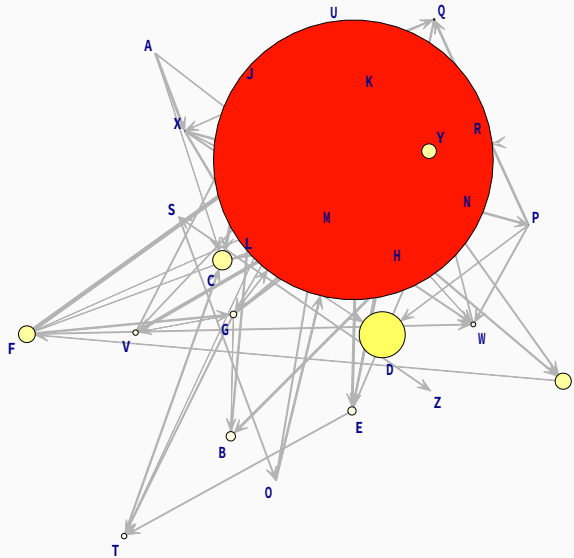
- without sink node escape
- initial: uniform dist.

$t = 9$



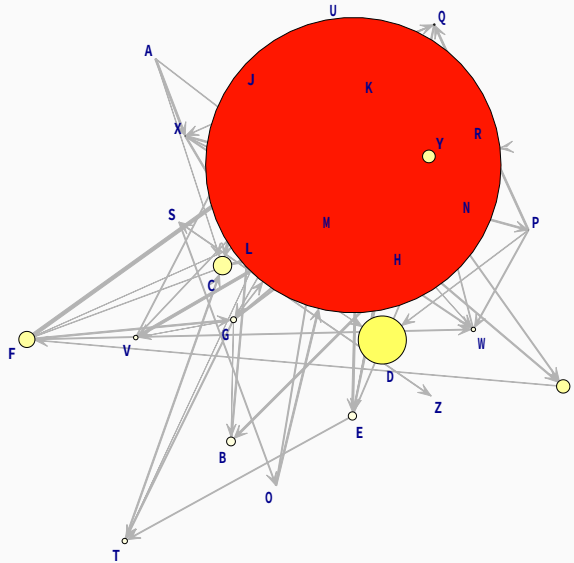
- without sink node escape
- initial: uniform dist.

$t = 10$



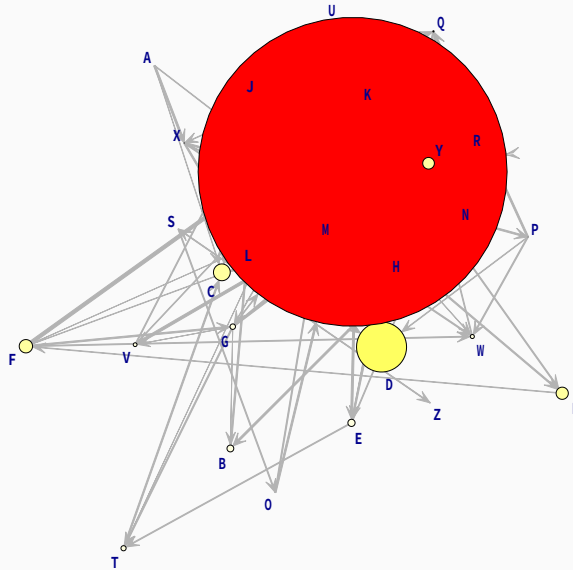
- without sink node escape
- initial: uniform dist.

t = 11



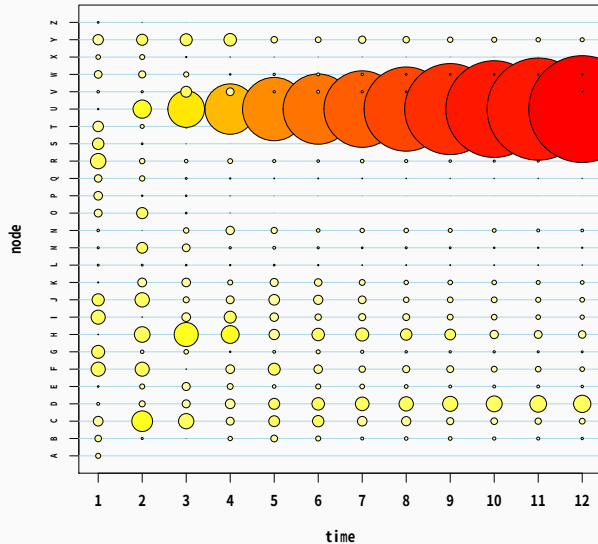
- without sink node escape
- initial: uniform dist.

$t = 12$



- without sink node escape
- initial: uniform dist.

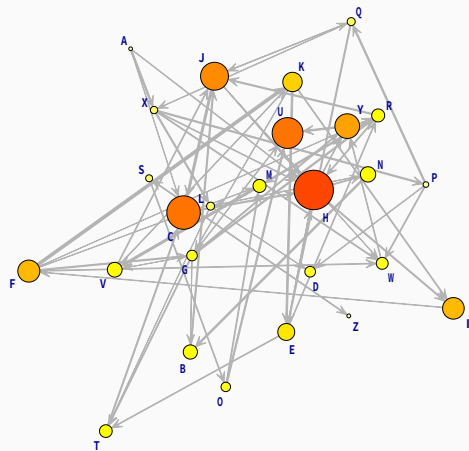
transient process



- without sink node escape
- initial: uniform dist.

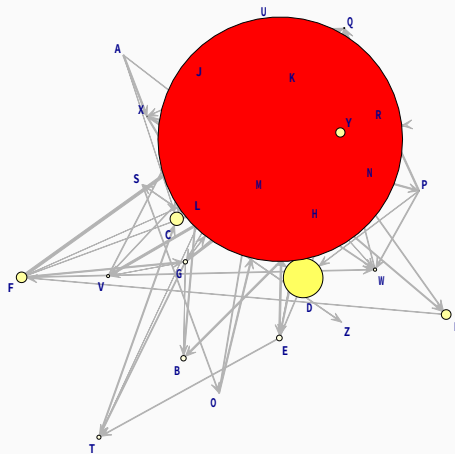
with sink node adjustment

$t = 12$



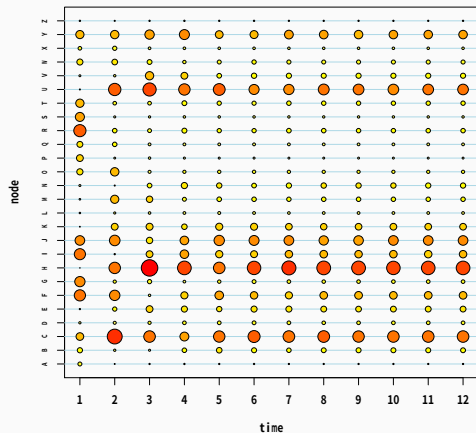
without sink node adjustment

$t = 12$



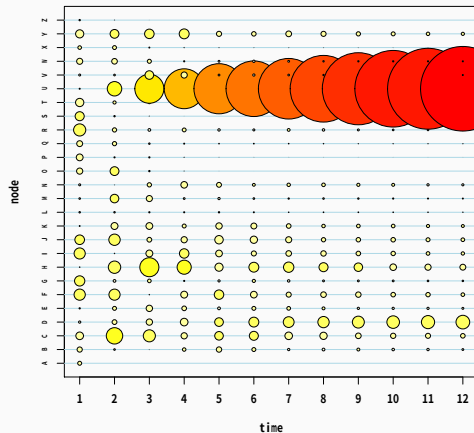
with sink node adjustment

transient process



without sink node adjustment

transient process



a simple and strong model of movements on directed graph:

- behavior model of selection from finite options
 - web surf model
 - purchase model of certain genres
 - transition model of audience ratings
 - customer share model of restaurants/coffee shops
- adjustment of transition matrix (sink node/ α)
 - out of stock or service
 - capricious or adventurous attempts
 - introduction from others

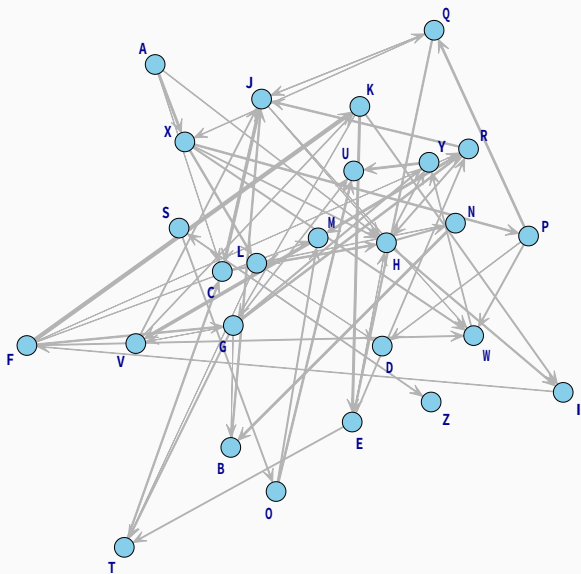
PROBLEM FORMULATION

non-stationary data on directed graphs:

- strength of edges slowly change in time
 - change of structure
 - change of stationary distribution
- model assumption:
 - frequent update (fast time scale; t)
e.g.: purchase every day
 - sparse observation (slow time scale; T)
e.g.: aggregate every week

observations are supposed to be on stationary distribution at current point

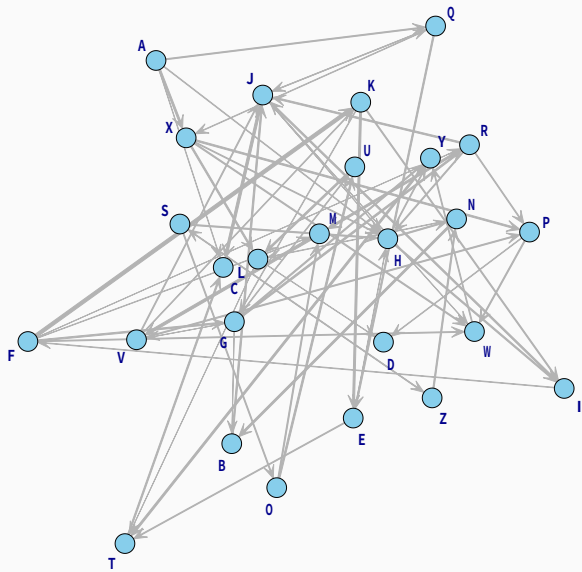
$T = 1$



change of structure

- 10% edges at random
- relatively small

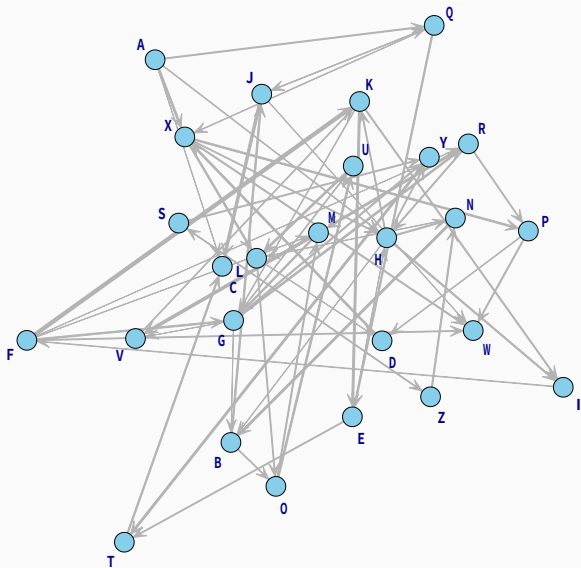
$T = 2$



change of structure

- 10% edges at random
- relatively small

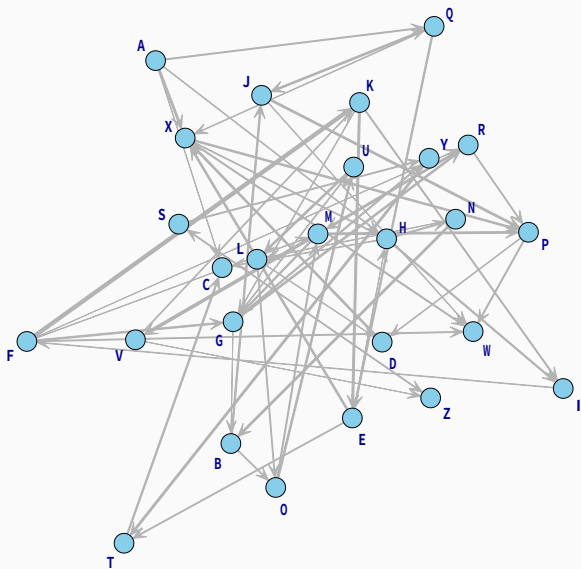
$T = 3$



change of structure

- 10% edges at random
- relatively small

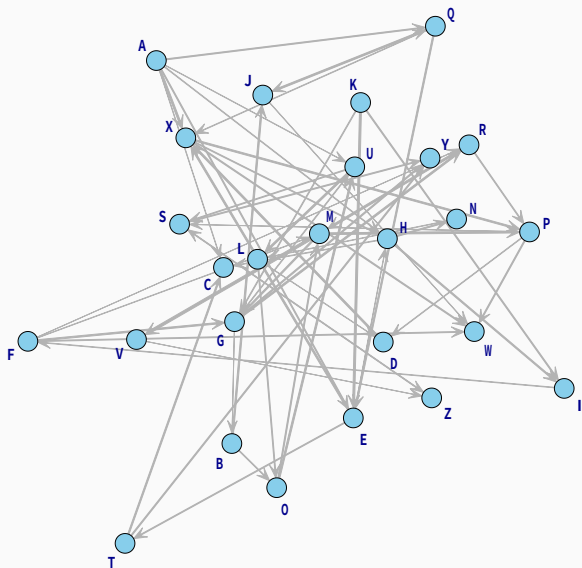
$T = 4$



change of structure

- 10% edges at random
- relatively small

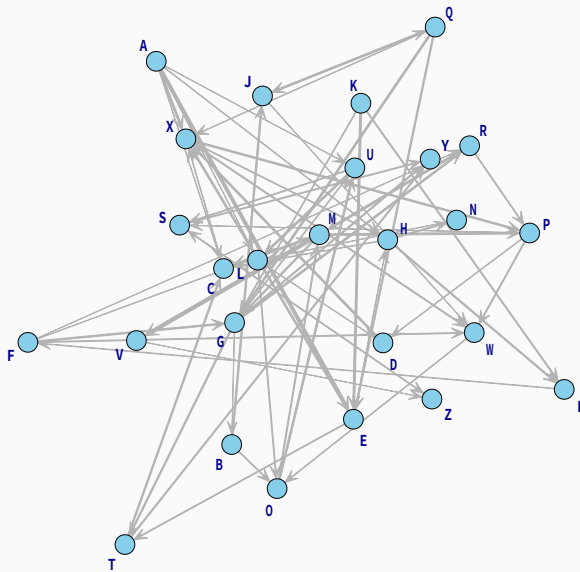
$T = 5$



change of structure

- 10% edges at random
- relatively small

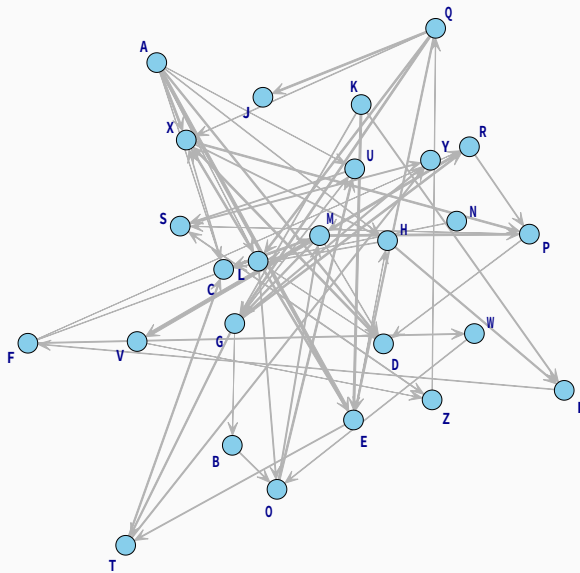
$T = 6$



change of structure

- 10% edges at random
- relatively small

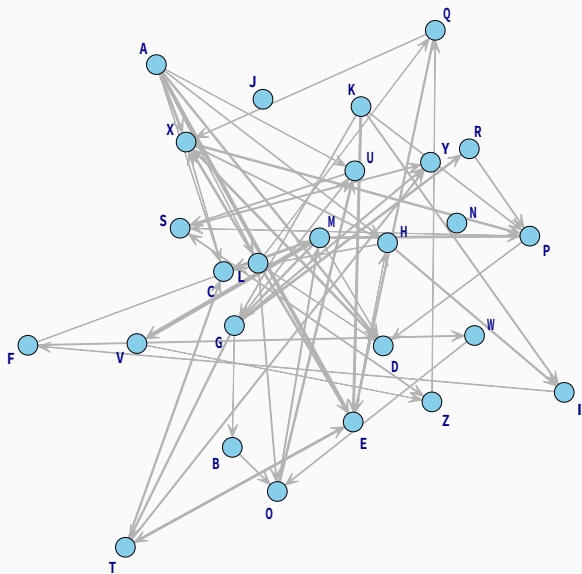
$T = 7$



change of structure

- 10% edges at random
- relatively small

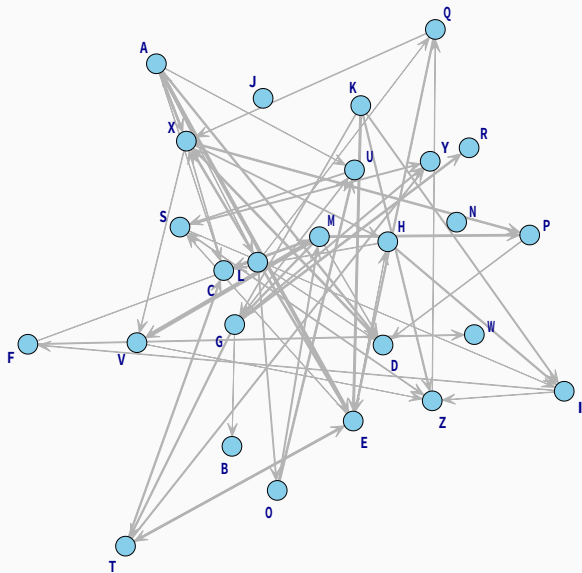
$T = 8$



change of structure

- 10% edges at random
- relatively small

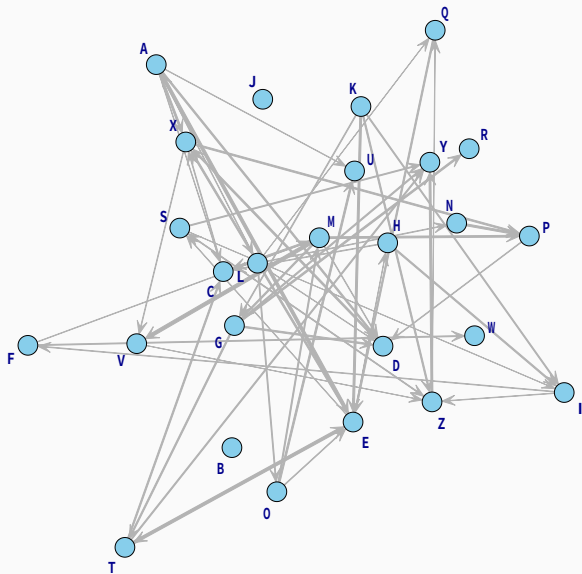
$T = 9$



change of structure

- 10% edges at random
- relatively small

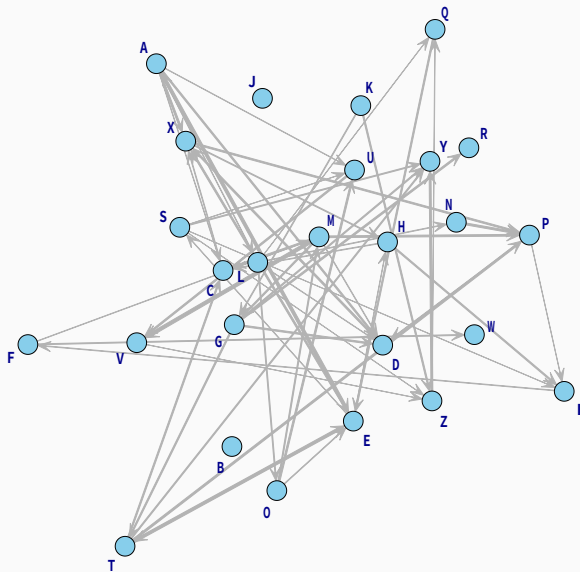
T = 10



change of structure

- 10% edges at random
- relatively small

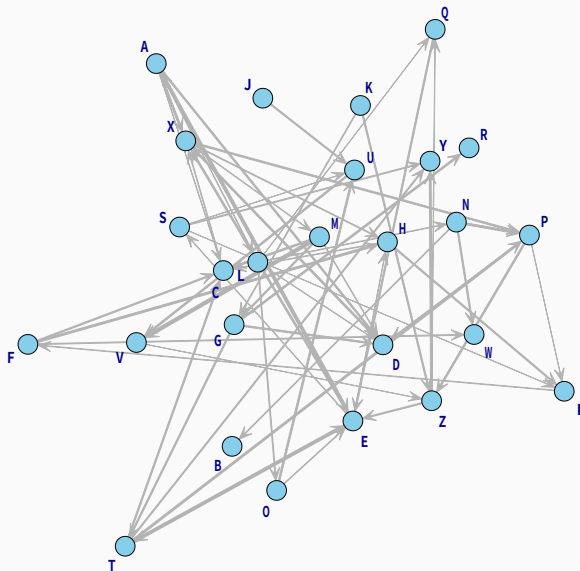
$T = 11$



change of structure

- 10% edges at random
- relatively small

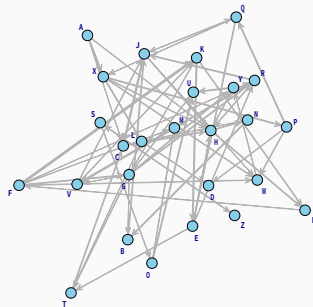
T = 12



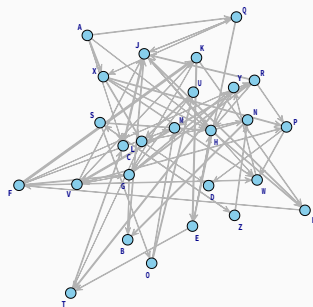
change of structure

- 10% edges at random
- relatively small

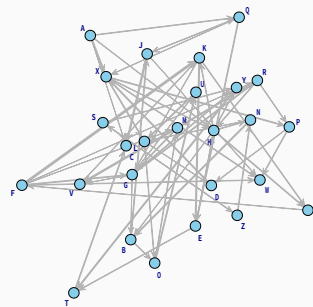
T = 1



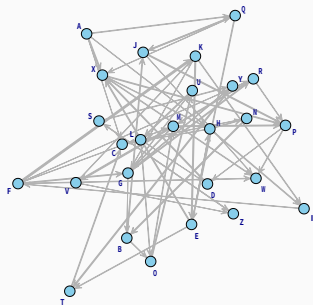
T = 2



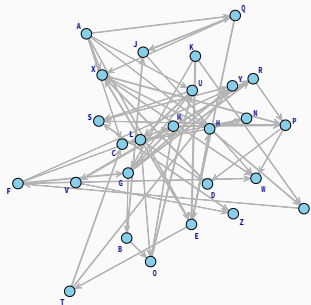
T = 3



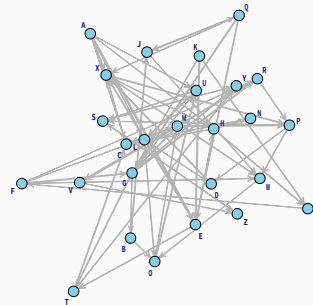
T = 4



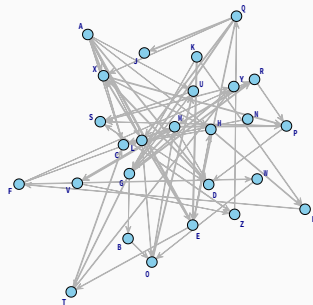
T = 5



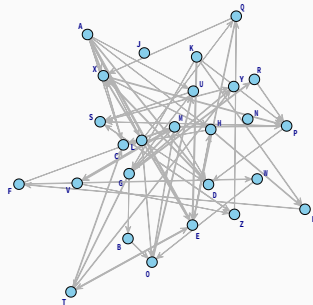
T = 6



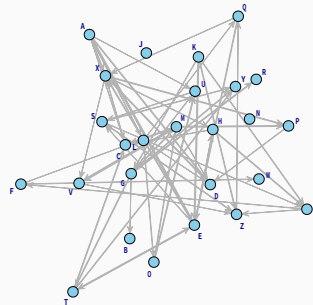
T = 7



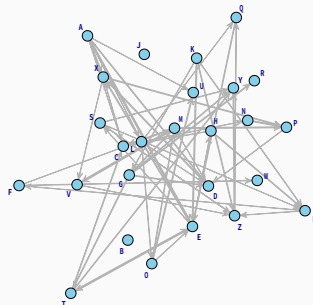
T = 8



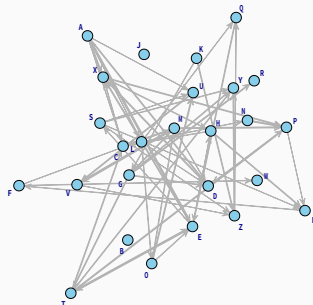
T = 9



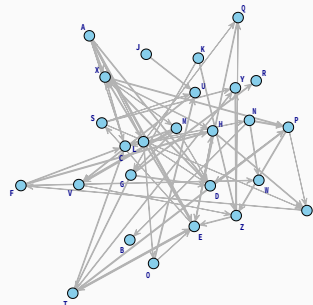
T = 10



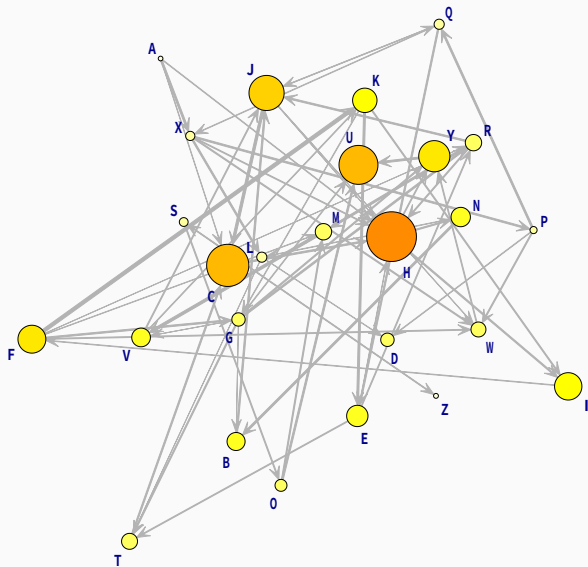
T = 11



T = 12



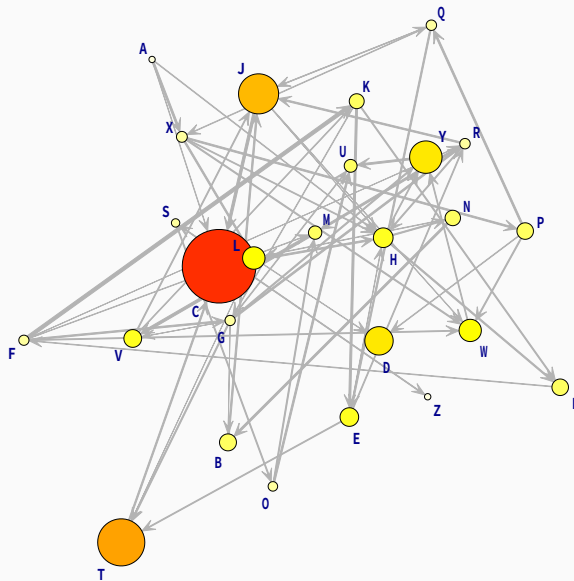
$T = 1$



change of stationary dist.

- 10% edges at random
- large effect

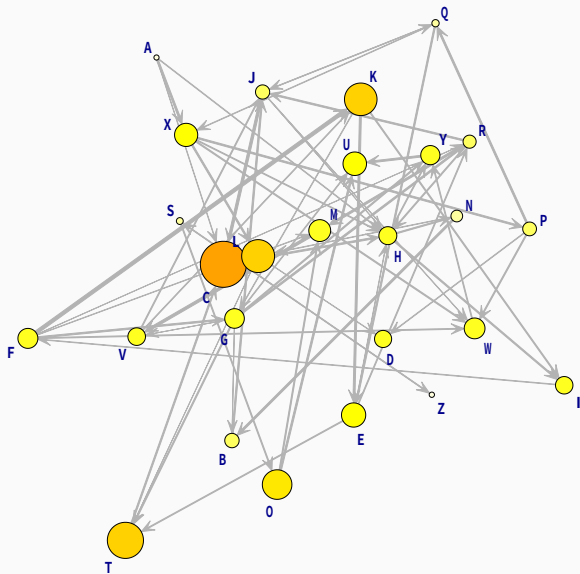
$T = 2$



change of stationary dist.

- 10% edges at random
- large effect

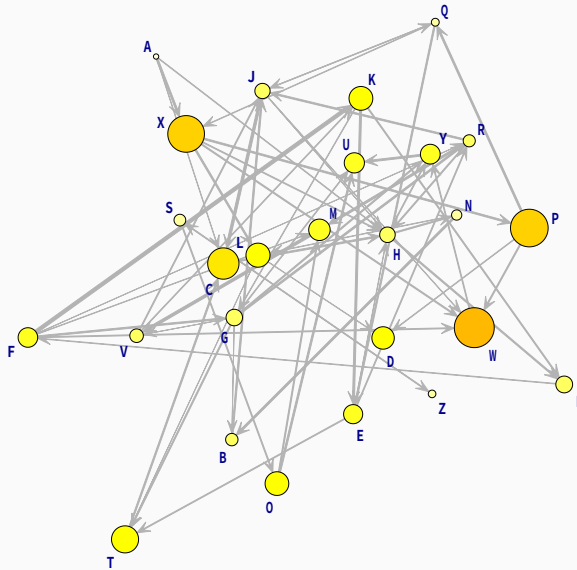
$T = 3$



change of stationary dist.

- 10% edges at random
- large effect

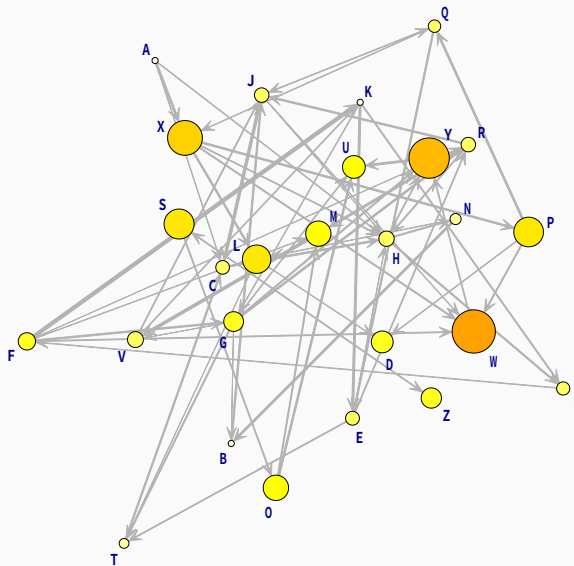
$T = 4$



change of stationary dist.

- 10% edges at random
- large effect

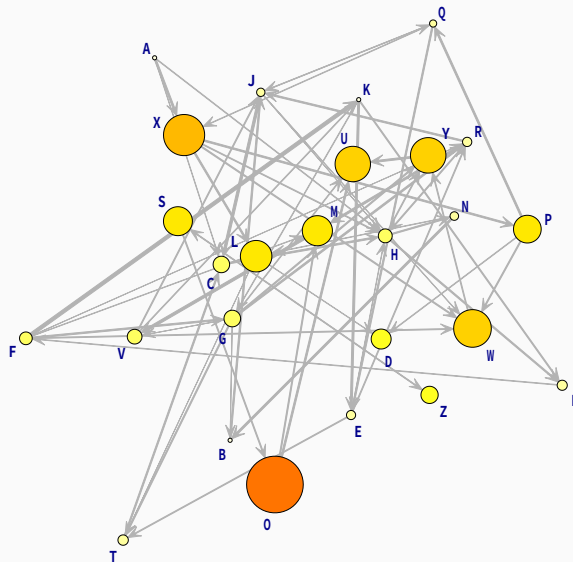
$T = 5$



change of stationary dist.

- 10% edges at random
- large effect

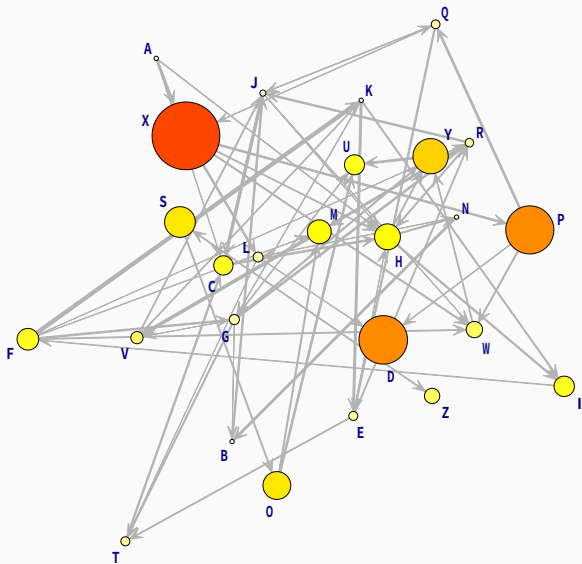
$T = 6$



change of stationary dist.

- 10% edges at random
- large effect

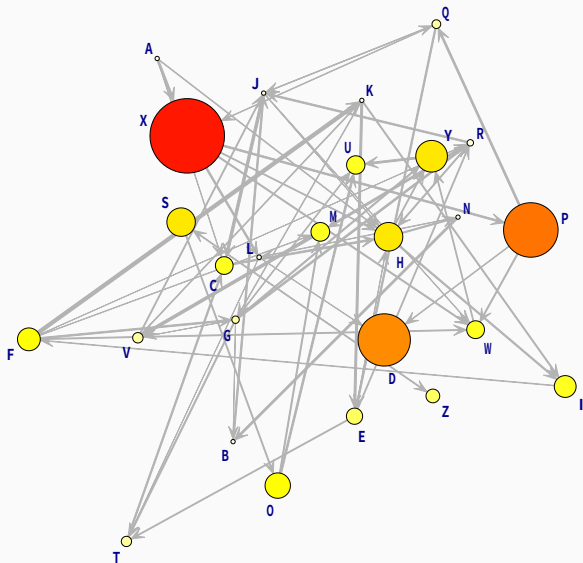
$T = 7$



change of stationary dist.

- 10% edges at random
- large effect

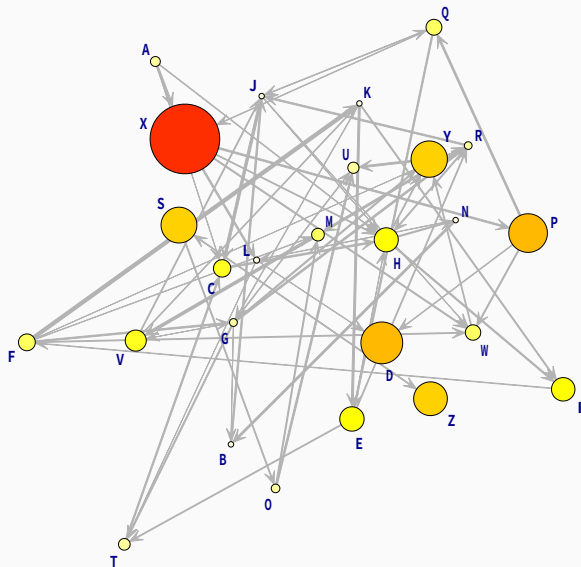
$T = 8$



change of stationary dist.

- 10% edges at random
- large effect

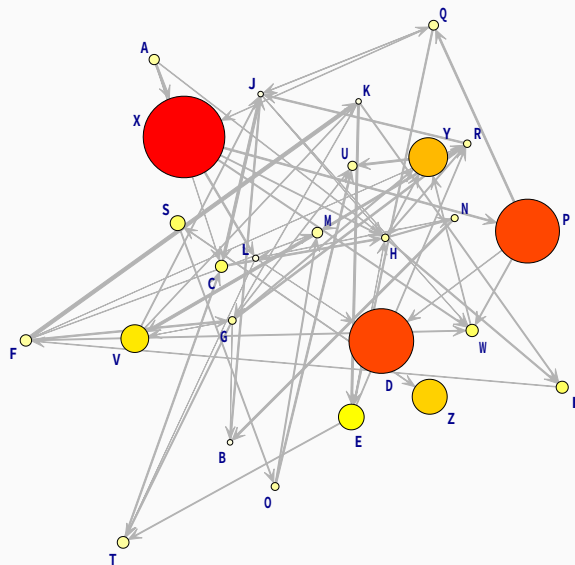
$T = 9$



change of stationary dist.

- 10% edges at random
- large effect

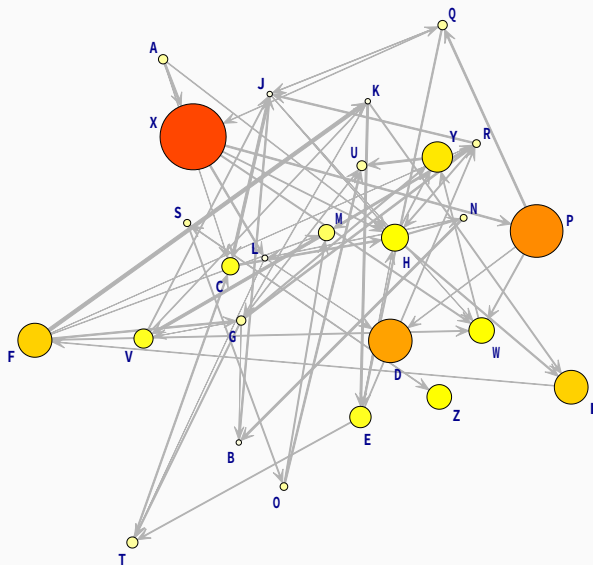
$T = 10$



change of stationary dist.

- 10% edges at random
- large effect

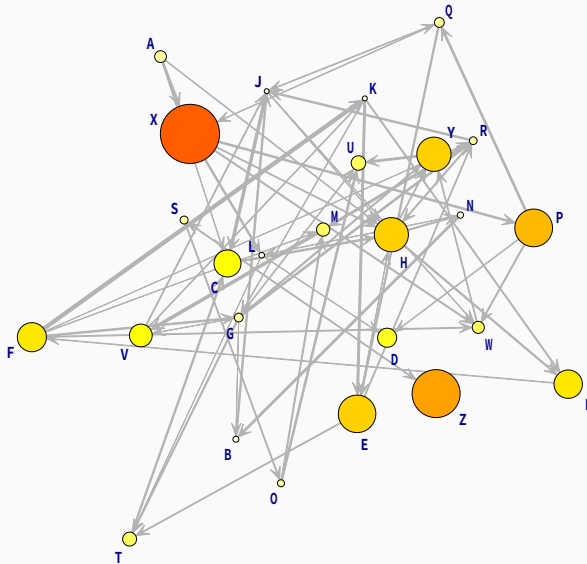
$T = 11$



change of stationary dist.

- 10% edges at random
- large effect

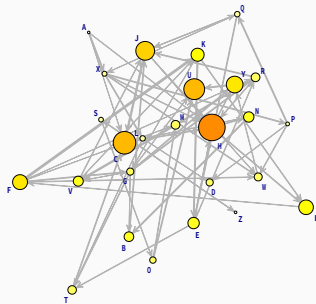
$T = 12$



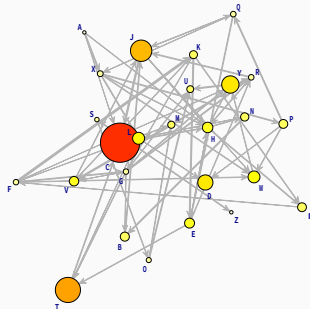
change of stationary dist.

- 10% edges at random
- large effect

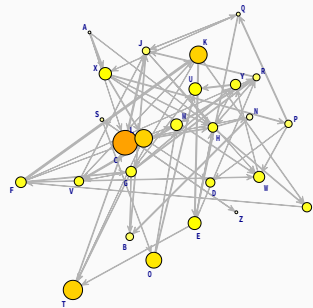
T = 1



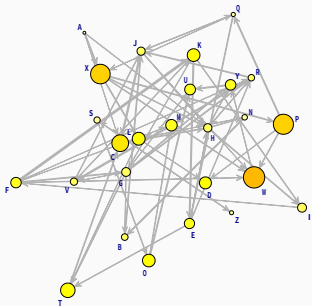
T = 2



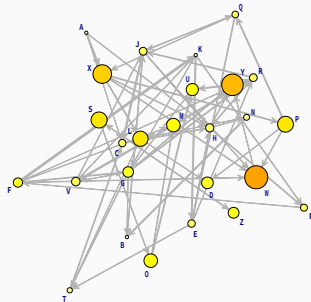
T = 3



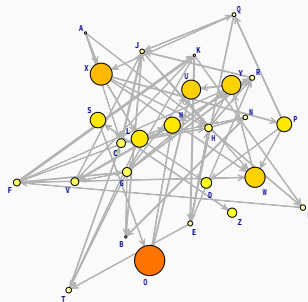
T = 4



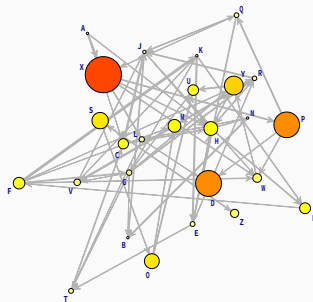
T = 5



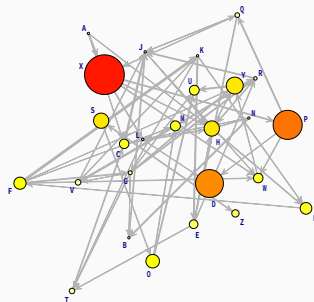
T = 6



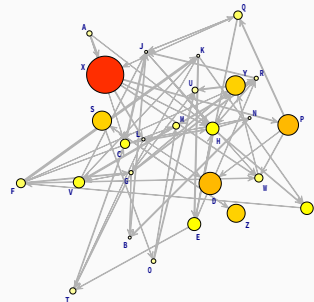
T = 7



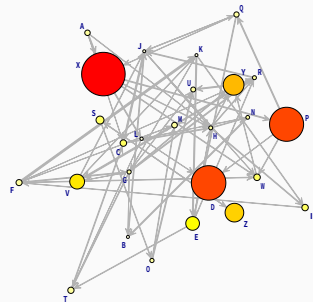
T = 8



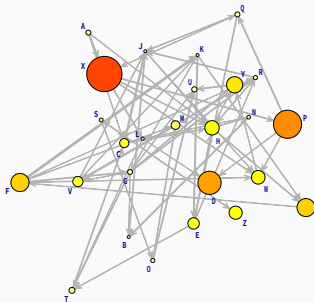
T = 9



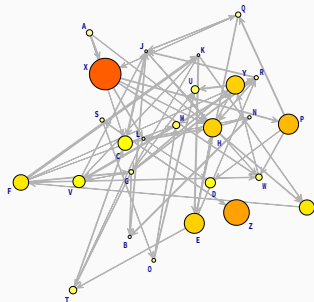
T = 10



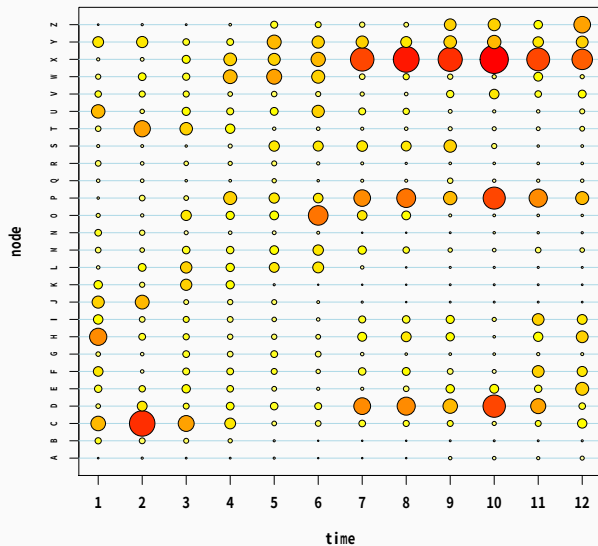
T = 11



T = 12



change of stationary states



change of stationary
distribution

- 10% edges at random
- large effect

for given series of stationary distributions $\{\pi_t\}$, estimate series of graph structures $\{G_t\}$.

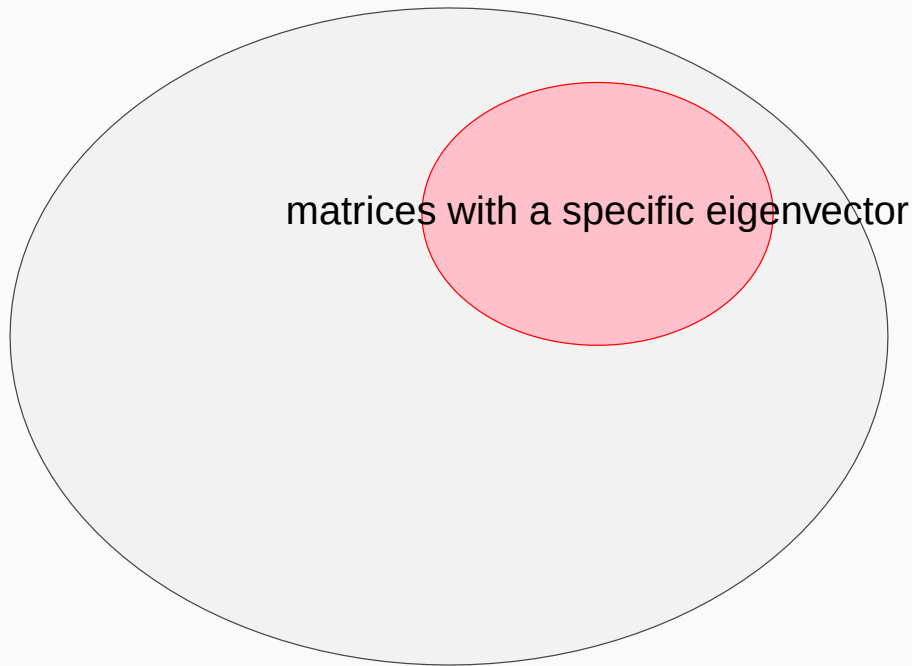
$$\text{minimize } L(\{G_t\}) \quad \text{subject to } \forall t, \pi_t^T G_t = \pi_t^T$$

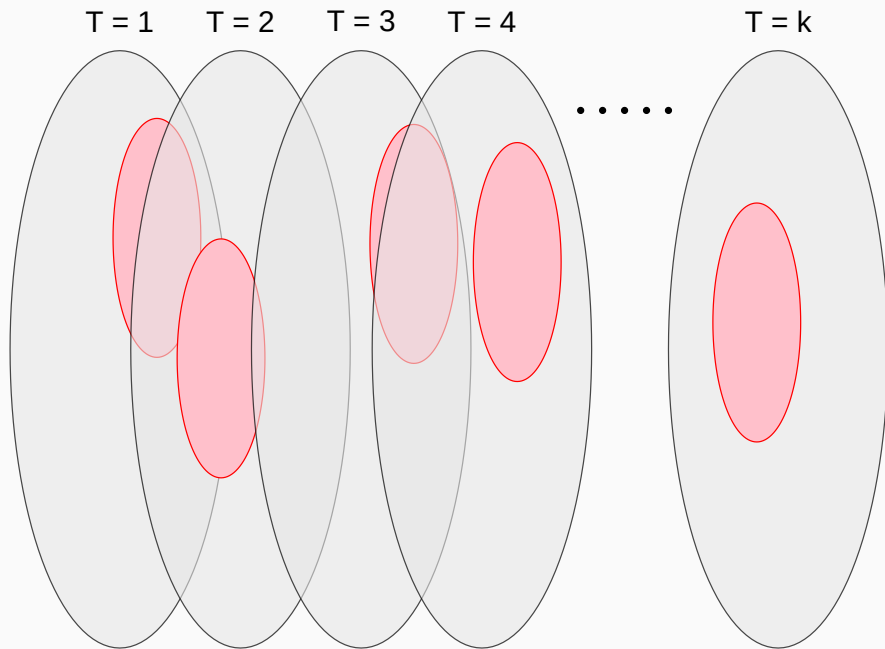
- infinitely many matrices have the same eigenvector
- the followings are needed:
 - assumptions on graph structures
 - assumptions of graph changes

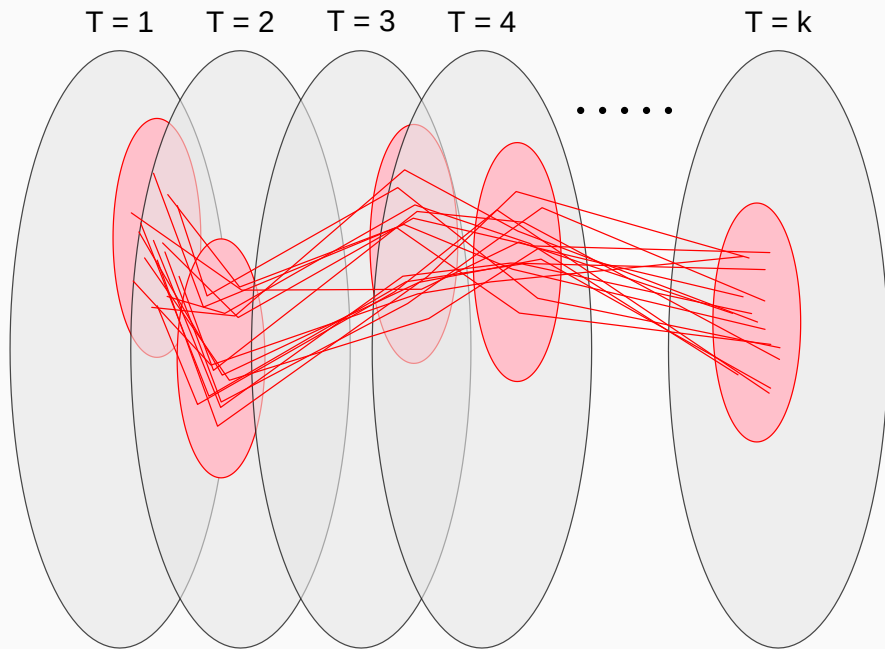
e.g. (fused lasso): for a certain sparse norm of matrix, $\|\cdot\|_S$,

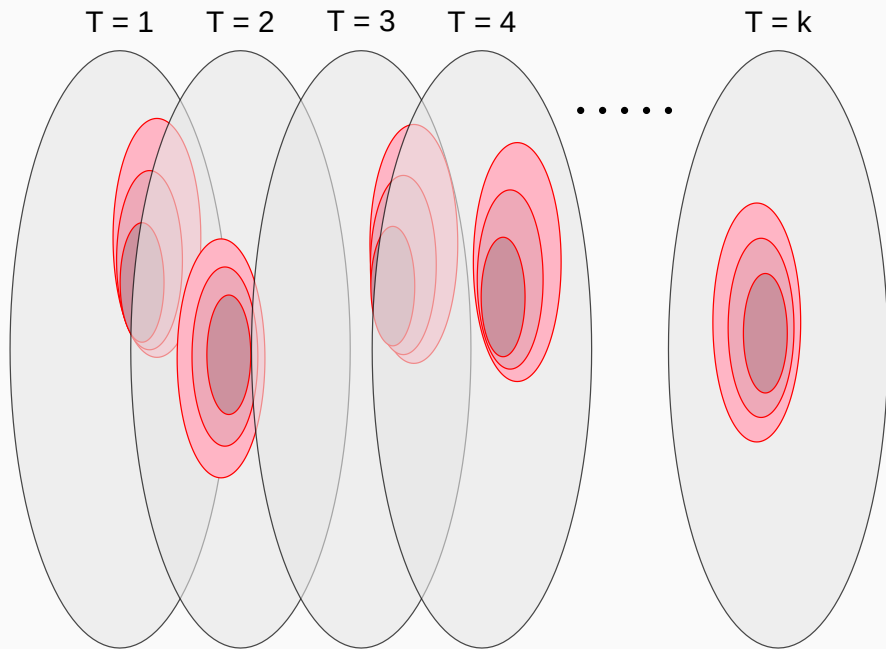
$$L(\{G_t\}) = \sum_t \|G_t\|_s + \sum_t \|G_{t+1} - G_t\|_s$$

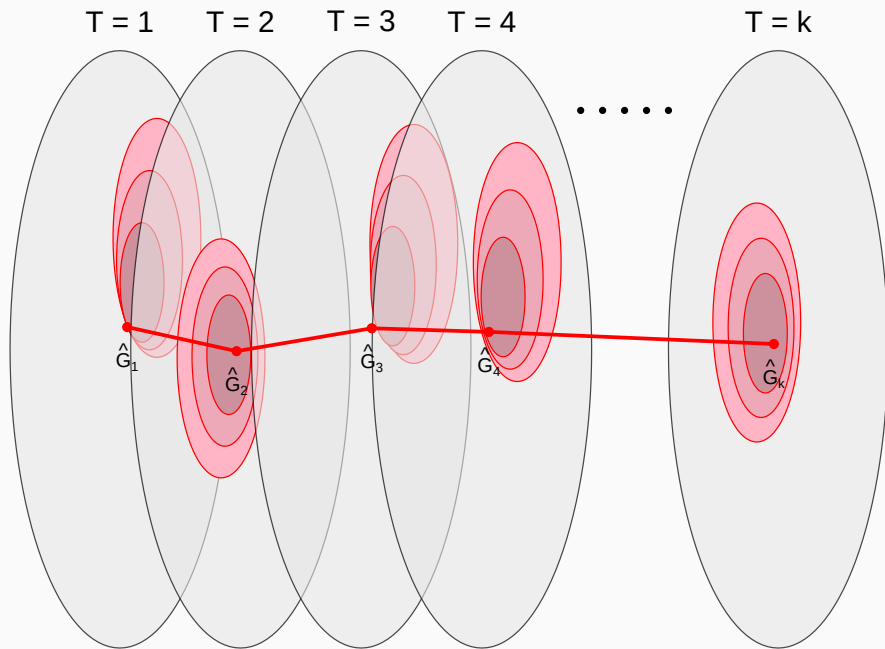
space of probability matrices











make matrices keep in a restricted subspace:

- eigenvector of matrix A to be $\mathbf{a}$ (unit vector):

$$A\mathbf{a} = \lambda\mathbf{a} + \mathbf{b} \Rightarrow A - \mathbf{b}\mathbf{a}^T \rightarrow A$$

- eigenvalue of eigenvector $\mathbf{a}$ to be μ :

$$A\mathbf{a} = \lambda\mathbf{a} \Rightarrow A + (\mu - \lambda)\mathbf{a}\mathbf{a}^T \rightarrow A$$

- A to be a probability matrix:

$$A\mathbf{e} = \mathbf{d} \Rightarrow \text{diag}(\mathbf{d})^{-1}A \rightarrow A$$

NUMERICAL EXAMPLES

Reference

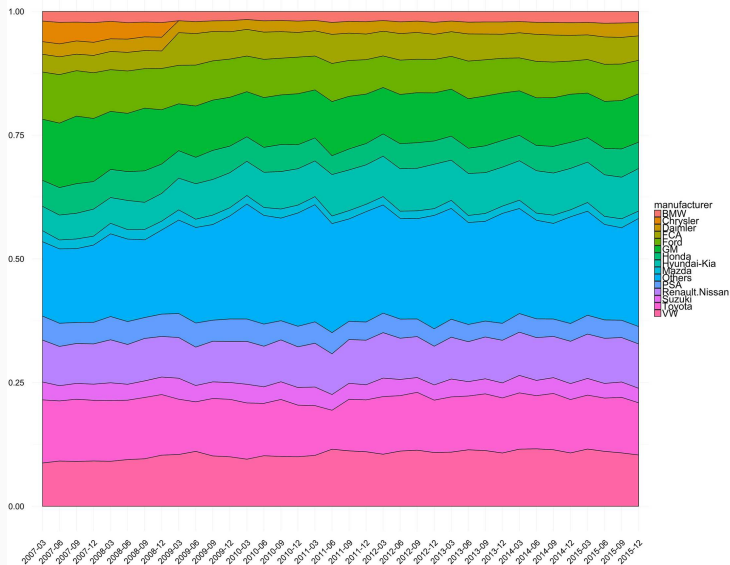
Chiba et al. "Time-Varying Transition Probability Matrix Estimation and Its Application to Brand Share Analysis"

- quarterly unit automobile sales data of manufacturers from 2007-1Q to 2015-4Q
- estimate transition paths and discuss the relation between social events and estimated results
- objective:

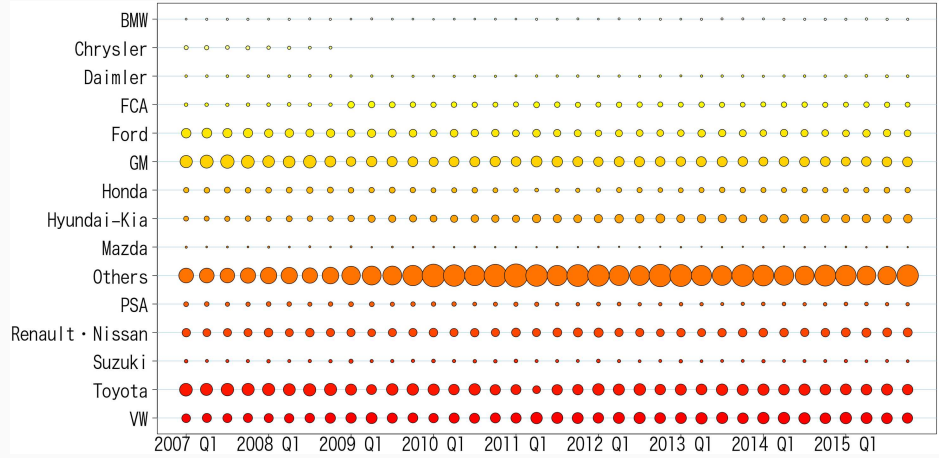
$$L(\{G_t\}) = \sum_t \|G_{t+1} - G_t\|_1$$

- optimization: simplex method with slack variables

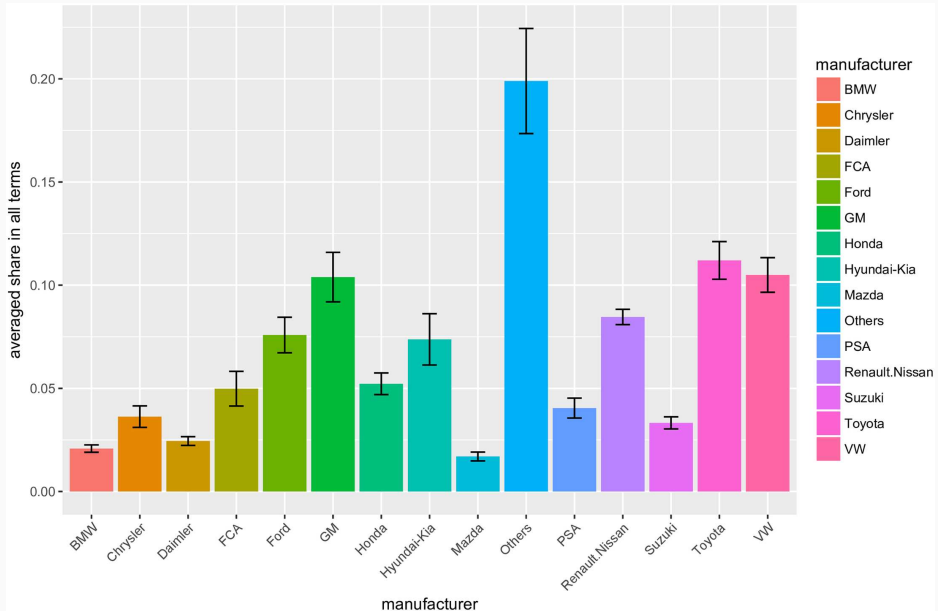
automobile sales for different manufactures

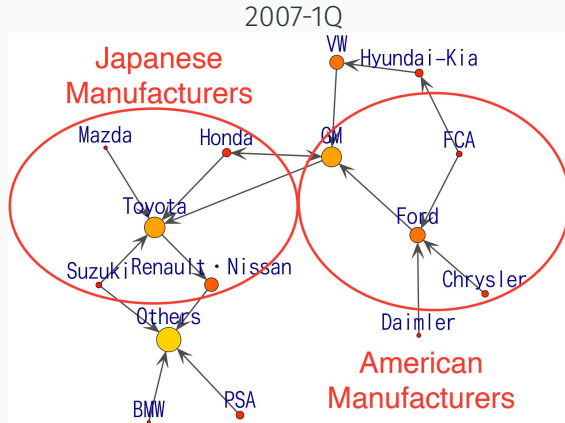


market share transition



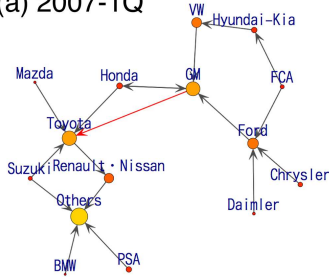
averages and standard deviations of sales shares



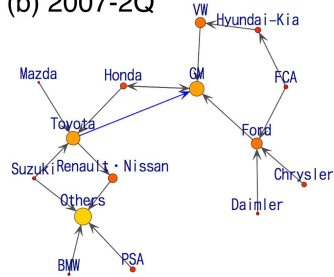


- remove minor edge below 0.24
- show market share with node size
- cf. GM and Honda are allied

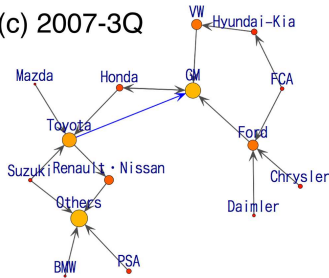
(a) 2007-1Q



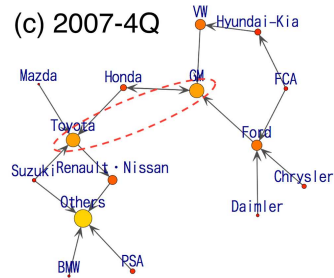
(b) 2007-2Q



(c) 2007-3Q

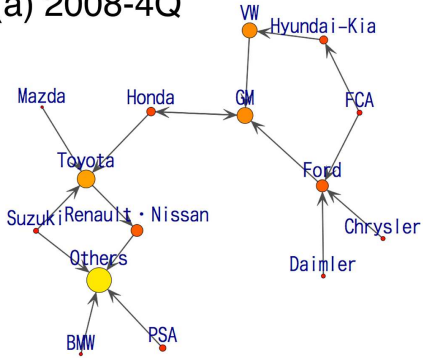


(c) 2007-4Q

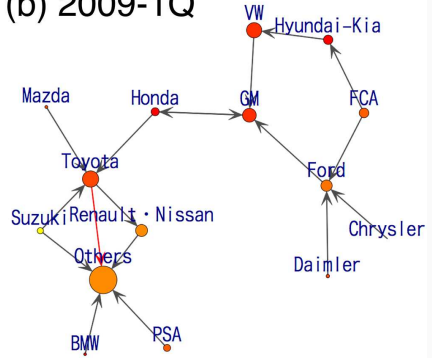


In March 2008, TOYOTA has become the world's top seller by beating GM

(a) 2008-4Q

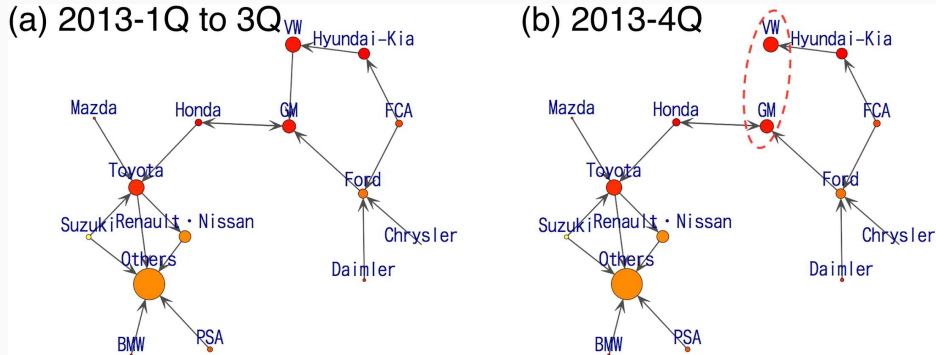


(b) 2009-1Q

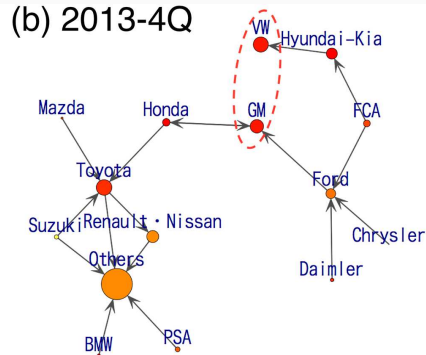


In 2009, TOYOTA launched a massive recall

(a) 2013-1Q to 3Q



(b) 2013-4Q



In 2013, VW beats GM in total sales amount to claim second position in the automobile industry

CONCLUSION

we presented the followings

- a model of transitions and stationary distributions
- a simple method for estimating transition matrices from a sequence of stationary distributions
- analysis of consumer transitions for sales share data without detailed recording of consumer transitions

further investigation would be devoted to

- other objectives and constraints to improve the accuracy of estimation and interpretability
- other probabilistic models for estimating changes in transitions

-  Chiba, Tomoaki et al. (Jan. 11, 2017). "Time-Varying Transition Probability Matrix Estimation and Its Application to Brand Share Analysis." In: *PLoS ONE* 12.1, e0169981. DOI: [10.1371/journal.pone.0169981](https://doi.org/10.1371/journal.pone.0169981).